

Figure 1: A diagram showing a cross-section of a structure with a curved top. The structure is labeled with various dimensions and parameters. The top part is a semi-circle with radius R . The base is a horizontal line. The structure is divided into two parts by a vertical line. The left part is labeled with dimensions a and b . The right part is labeled with dimensions c and d . The total width is $2R$. The height of the structure is H . The diagram is labeled with various parameters and dimensions.

Figure 2: A diagram showing a cross-section of a structure with a curved top. The structure is labeled with various dimensions and parameters. The top part is a semi-circle with radius R . The base is a horizontal line. The structure is divided into two parts by a vertical line. The left part is labeled with dimensions a and b . The right part is labeled with dimensions c and d . The total width is $2R$. The height of the structure is H . The diagram is labeled with various parameters and dimensions.

Figure 3: A diagram showing a cross-section of a structure with a curved top. The structure is labeled with various dimensions and parameters. The top part is a semi-circle with radius R . The base is a horizontal line. The structure is divided into two parts by a vertical line. The left part is labeled with dimensions a and b . The right part is labeled with dimensions c and d . The total width is $2R$. The height of the structure is H . The diagram is labeled with various parameters and dimensions.

Figure 4: A diagram showing a cross-section of a structure with a curved top. The structure is labeled with various dimensions and parameters. The top part is a semi-circle with radius R . The base is a horizontal line. The structure is divided into two parts by a vertical line. The left part is labeled with dimensions a and b . The right part is labeled with dimensions c and d . The total width is $2R$. The height of the structure is H . The diagram is labeled with various parameters and dimensions.

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1. **Identify the main components of the system.** The system consists of a central processing unit (CPU), memory, and input/output devices.

1. **QUESTION** (100 points) (100 points)

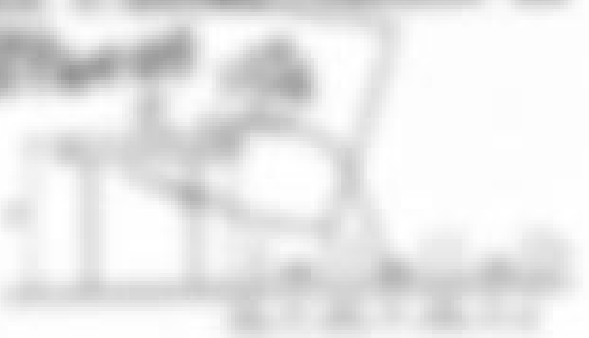
1. The following information is for the purpose of the
training. It is not to be used for any other purpose.
The information is for the purpose of the training.
The information is for the purpose of the training.



2. The following information is for the purpose of the
training. It is not to be used for any other purpose.
The information is for the purpose of the training.
The information is for the purpose of the training.

3. The following information is for the purpose of the
training. It is not to be used for any other purpose.
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1. A rectangular plate of width b and height h is subjected to a uniformly distributed load q acting downwards. The plate is supported by a pin support at the bottom left corner and a roller support at the bottom right corner. Determine the reaction forces at the supports.

2. A triangular plate of base b and height h is subjected to a uniformly distributed load q acting downwards. The plate is supported by a pin support at the bottom left corner and a roller support at the bottom right corner. Determine the reaction forces at the supports.

- q (N/m)
- b (m)
- h (m)
- R_A (N)
- R_B (N)
- R_C (N)
- R_D (N)
- R_E (N)
- R_F (N)
- R_G (N)
- R_H (N)
- R_I (N)
- R_J (N)
- R_K (N)
- R_L (N)
- R_M (N)
- R_N (N)
- R_O (N)
- R_P (N)
- R_Q (N)
- R_R (N)
- R_S (N)
- R_T (N)
- R_U (N)
- R_V (N)
- R_W (N)
- R_X (N)
- R_Y (N)
- R_Z (N)

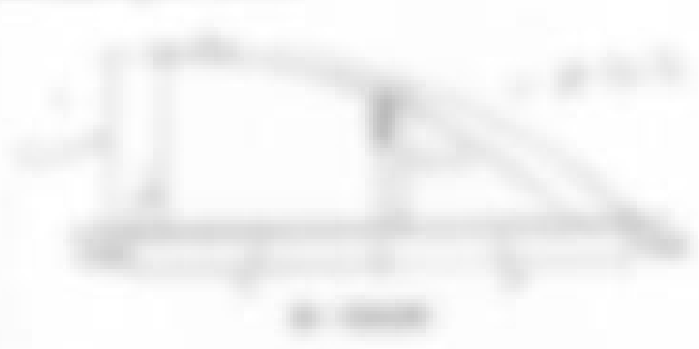
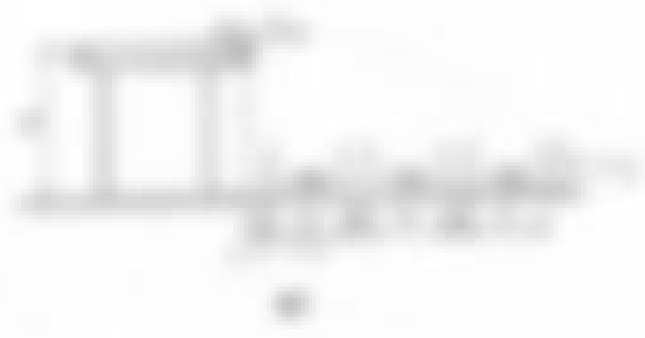


Figure 1

3. A rectangular plate of width b and height h is subjected to a uniformly distributed load q acting downwards. The plate is supported by a pin support at the bottom left corner and a roller support at the bottom right corner. Determine the reaction forces at the supports.

- q (N/m)
- b (m)
- h (m)
- R_A (N)
- R_B (N)
- R_C (N)
- R_D (N)
- R_E (N)
- R_F (N)
- R_G (N)
- R_H (N)
- R_I (N)
- R_J (N)
- R_K (N)
- R_L (N)
- R_M (N)
- R_N (N)
- R_O (N)
- R_P (N)
- R_Q (N)
- R_R (N)
- R_S (N)
- R_T (N)
- R_U (N)
- R_V (N)
- R_W (N)
- R_X (N)
- R_Y (N)
- R_Z (N)



1. A rectangular block of material is shown in the figure. The block is subjected to a uniform pressure p acting on all its faces. The block is divided into two equal parts by a vertical plane. The pressure p is acting on the top and bottom faces of the block. The pressure p is acting on the left and right faces of the block. The pressure p is acting on the front and back faces of the block. The pressure p is acting on the top and bottom faces of the block. The pressure p is acting on the left and right faces of the block. The pressure p is acting on the front and back faces of the block.

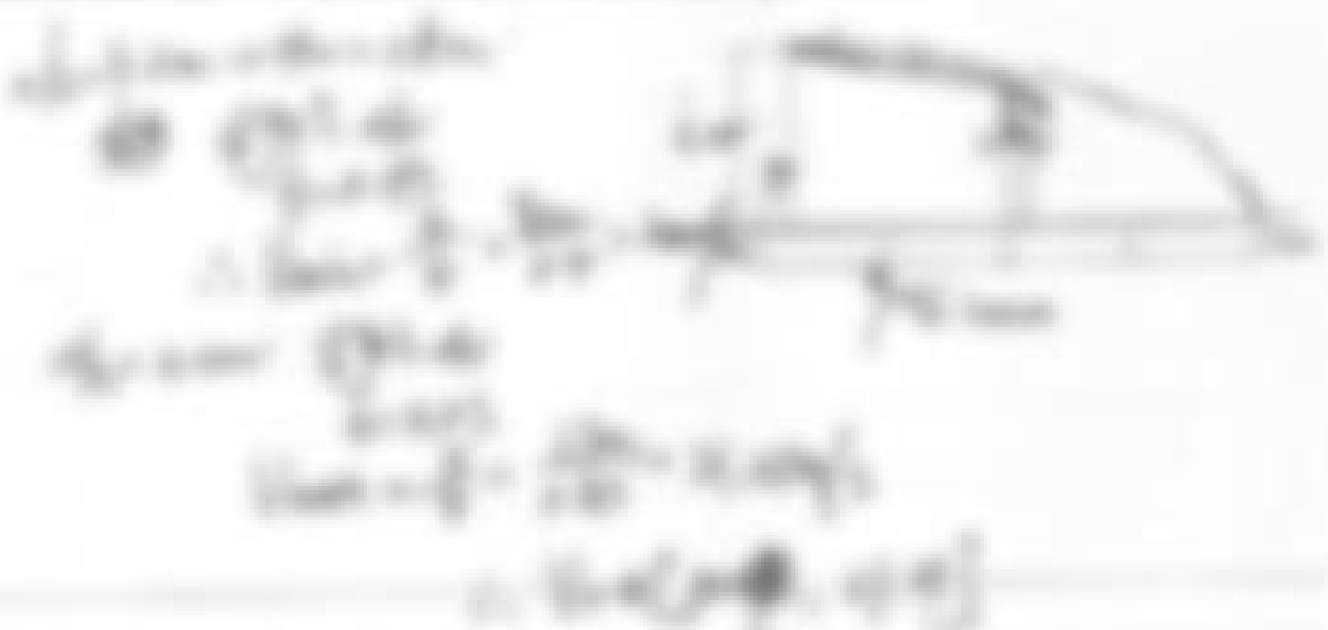


2. A rectangular block of material is shown in the figure. The block is subjected to a uniform pressure p acting on all its faces. The block is divided into two equal parts by a vertical plane. The pressure p is acting on the top and bottom faces of the block. The pressure p is acting on the left and right faces of the block. The pressure p is acting on the front and back faces of the block. The pressure p is acting on the top and bottom faces of the block. The pressure p is acting on the left and right faces of the block. The pressure p is acting on the front and back faces of the block.

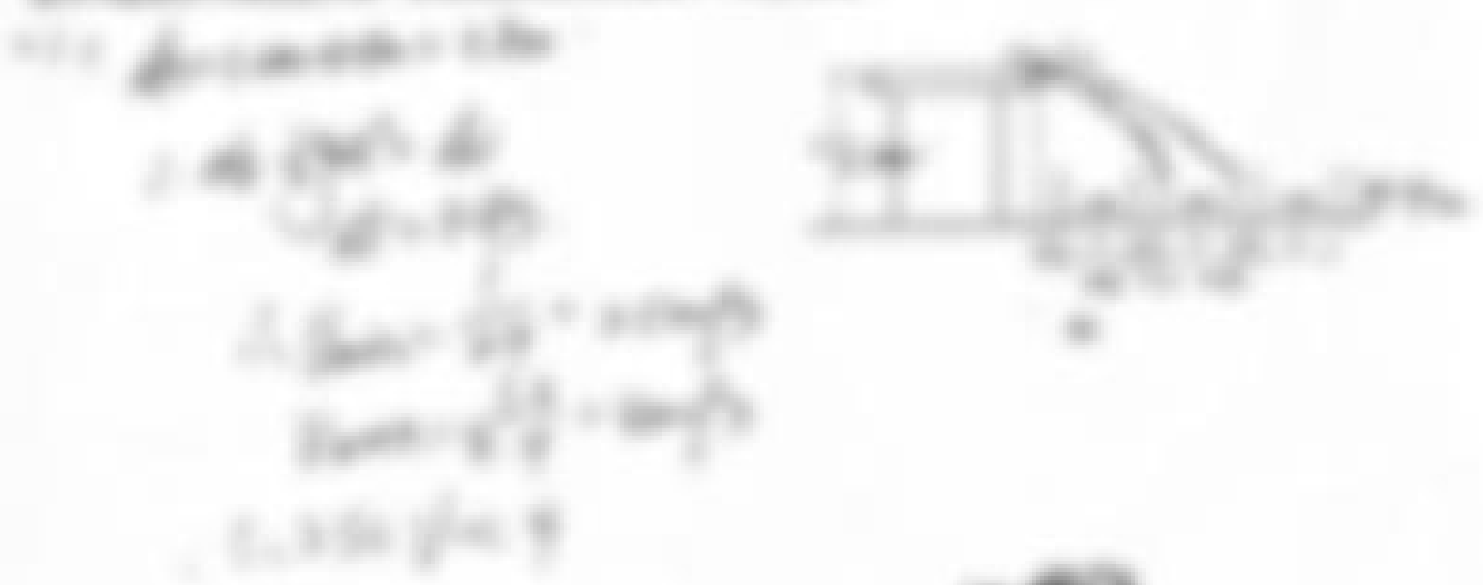
- a. A rectangular block of material is shown in the figure. The block is subjected to a uniform pressure p acting on all its faces. The block is divided into two equal parts by a vertical plane. The pressure p is acting on the top and bottom faces of the block. The pressure p is acting on the left and right faces of the block. The pressure p is acting on the front and back faces of the block.
- b. A rectangular block of material is shown in the figure. The block is subjected to a uniform pressure p acting on all its faces. The block is divided into two equal parts by a vertical plane. The pressure p is acting on the top and bottom faces of the block. The pressure p is acting on the left and right faces of the block. The pressure p is acting on the front and back faces of the block.



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[illegible]

1. **QUESTION** (100 points)



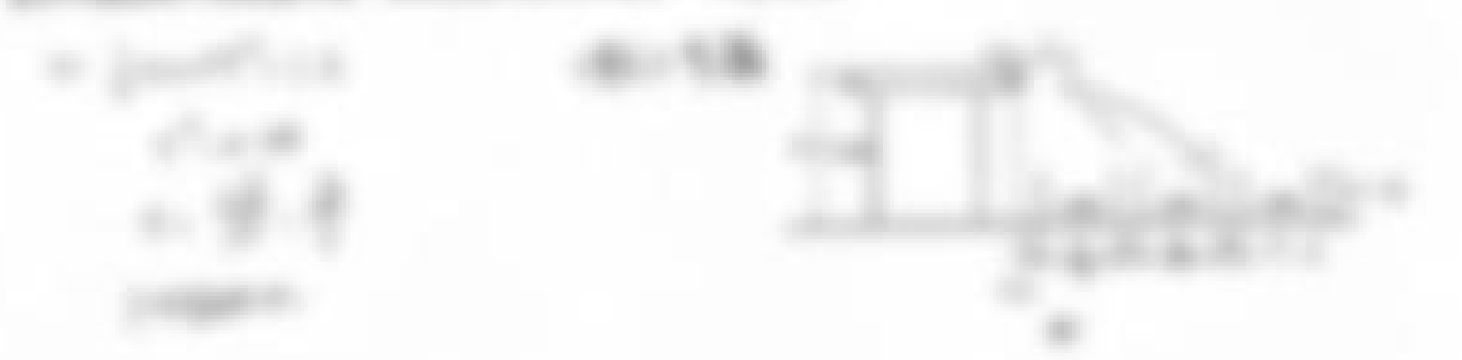
1992

1. The following information is taken from the financial statements of a company for the year ended 31 December 2018. The company has a number of subsidiaries and is a member of a group. The company's financial statements are prepared in accordance with the requirements of the Companies Act 2006. The company's financial statements are prepared in accordance with the requirements of the Companies Act 2006.



2. The following information is taken from the financial statements of a company for the year ended 31 December 2018. The company has a number of subsidiaries and is a member of a group. The company's financial statements are prepared in accordance with the requirements of the Companies Act 2006. The company's financial statements are prepared in accordance with the requirements of the Companies Act 2006.

- a. The company has a number of subsidiaries and is a member of a group.
- b. The company's financial statements are prepared in accordance with the requirements of the Companies Act 2006.



[illegible]

1. **RESEARCH DESIGN**—This study was a descriptive, cross-sectional survey of the prevalence of self-reported depression among a convenience sample of 1000 U.S. adults. The survey was conducted between January and February 2001. The survey was conducted by a research firm (Survey Research Center, Inc., Ann Arbor, Michigan) using a computer-assisted telephone interview (CATI) system. The survey was conducted in English and Spanish. The survey was conducted in English and Spanish. The survey was conducted in English and Spanish.

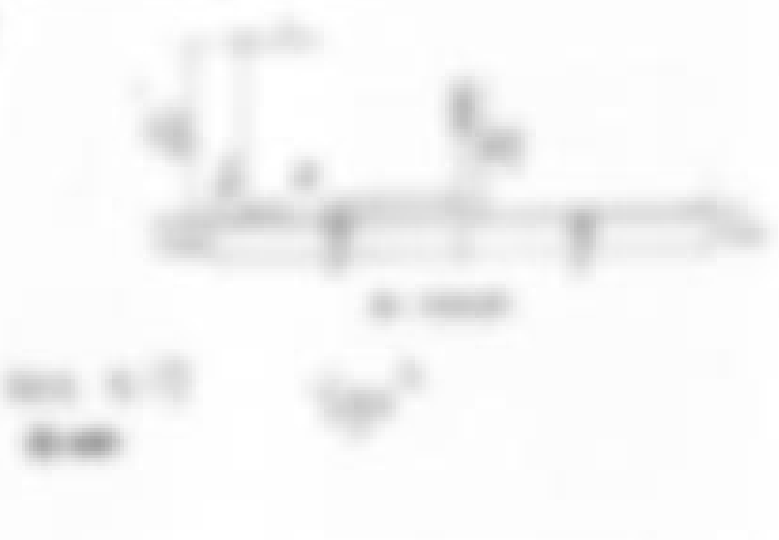
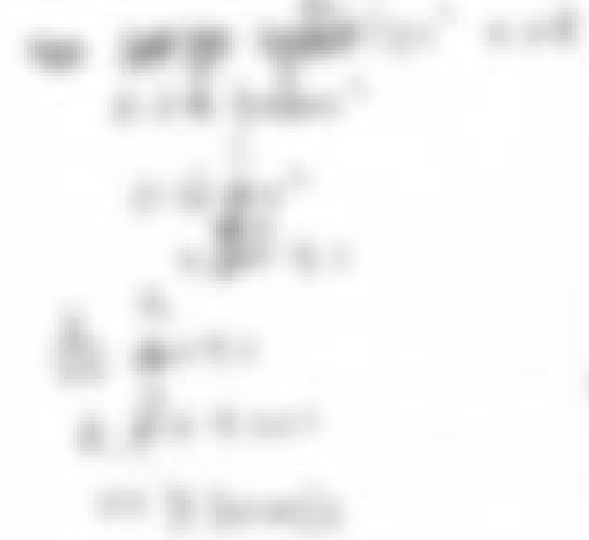
1. *Journal of Management Education*, 2000, 24(1), 1-10.

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Keywords: child sexual abuse; disclosure; social support

1. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom edge. The block is on the verge of tipping over the right edge. Find the magnitude of the force F in terms of m , h , and y .

2. A uniform rod of length L and mass M is pivoted at the left end. A force F is applied perpendicular to the rod at the right end. Find the magnitude of the force F required to keep the rod horizontal.



3. A uniform rod of length L and mass M is pivoted at the left end. A force F is applied perpendicular to the rod at the right end. Find the magnitude of the force F required to keep the rod horizontal.

- (a) Find the magnitude of the force F required to keep the rod horizontal.
- (b) Find the magnitude of the force F required to keep the rod horizontal.



1. **PROBABILITY THEORY**
DEFINITION: PROBABILITY IS THE RATIO OF THE NUMBER OF OUTCOMES FAVORABLE TO AN EVENT TO THE TOTAL NUMBER OF OUTCOMES.

PROBABILITY DISTRIBUTION



Let X be a random variable. The probability distribution of X is the set of probabilities $P(X=x_i)$ for all possible values x_i of X . The sum of all probabilities is 1.0.

2. **STATISTICAL INFERENCE**
DEFINITION: STATISTICAL INFERENCE IS THE PROCESS OF DRAWING INFERENCES ABOUT THE POPULATION FROM THE SAMPLE DATA.

1. **ESTIMATION**
2. **TESTING**



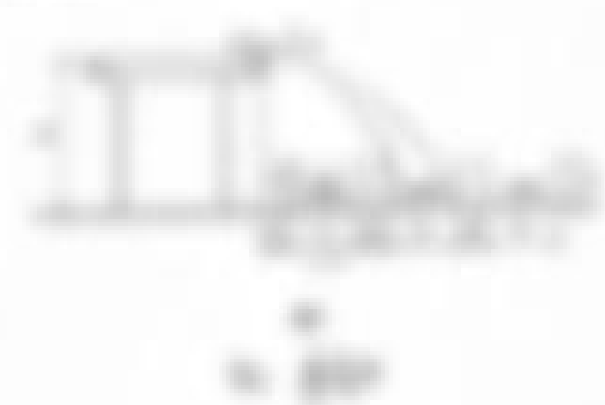
1. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied at the top right corner, pulling the block to the right. The force F is directed at an angle θ to the horizontal. The block is on the verge of tipping over its right edge. Find the magnitude of the force F in terms of m , h , and θ .

Given: m , h , θ
Find: F
Diagram: A rectangular block of mass m and height h is shown on a horizontal surface. A force F is applied at the top right corner, pulling the block to the right. The force F is directed at an angle θ to the horizontal. The block is on the verge of tipping over its right edge.



2. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied at the top left corner, pulling the block to the right. The force F is directed at an angle θ to the horizontal. The block is on the verge of tipping over its left edge. Find the magnitude of the force F in terms of m , h , and θ .

Given: m , h , θ
Find: F
Diagram: A rectangular block of mass m and height h is shown on a horizontal surface. A force F is applied at the top left corner, pulling the block to the right. The force F is directed at an angle θ to the horizontal. The block is on the verge of tipping over its left edge.

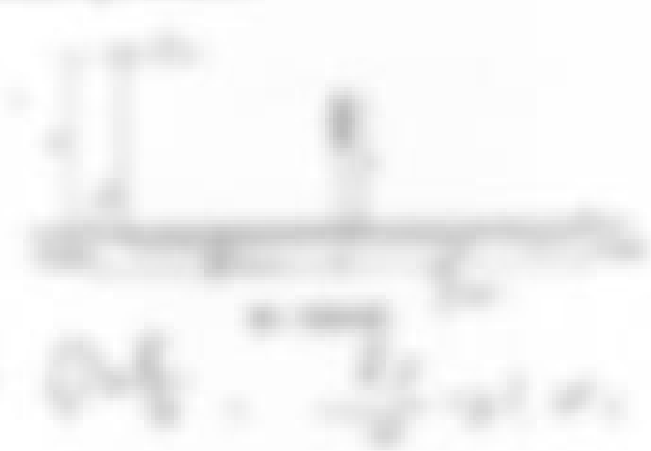


Answer: $F = \frac{mgh}{2 \sin \theta}$

1. **အထွေထွေ အချက်အလက်** (General Information)
 အမည်: **အောင်ကျော်** (Name: Aung Mye Kywe)
 နံပါတ်: **၁၂၃၄၅** (Number: 12345)

2. **အချက်အလက်** (Details)

အမည်
 နံပါတ်
 နေရာ
 အခြား



အခြားအချက်အလက်

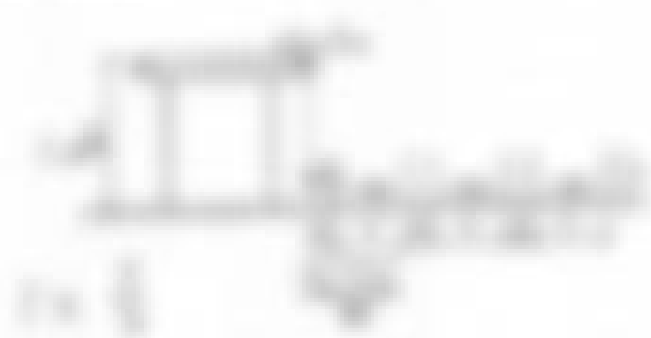
အခြားအချက်အလက်

3. **အချက်အလက်** (Details)
 အမည်: **အောင်ကျော်** (Name: Aung Mye Kywe)
 နံပါတ်: **၁၂၃၄၅** (Number: 12345)
 နေရာ: **၁၂၃၄၅** (Location: 12345)
 အခြားအချက်အလက်: **၁၂၃၄၅** (Other details: 12345)

4. **အချက်အလက်** (Details)
 အမည်: **အောင်ကျော်** (Name: Aung Mye Kywe)
 နံပါတ်: **၁၂၃၄၅** (Number: 12345)
 နေရာ: **၁၂၃၄၅** (Location: 12345)
 အခြားအချက်အလက်: **၁၂၃၄၅** (Other details: 12345)

5. **အချက်အလက်** (Details)

အမည်
 နံပါတ်
 နေရာ



အခြားအချက်အလက်
 အခြားအချက်အလက်

6. **အချက်အလက်** (Details)
 အမည်: **အောင်ကျော်** (Name: Aung Mye Kywe)
 နံပါတ်: **၁၂၃၄၅** (Number: 12345)
 နေရာ: **၁၂၃၄၅** (Location: 12345)
 အခြားအချက်အလက်: **၁၂၃၄၅** (Other details: 12345)

7. **အချက်အလက်** (Details)
 အမည်: **အောင်ကျော်** (Name: Aung Mye Kywe)
 နံပါတ်: **၁၂၃၄၅** (Number: 12345)
 နေရာ: **၁၂၃၄၅** (Location: 12345)
 အခြားအချက်အလက်: **၁၂၃၄၅** (Other details: 12345)

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Figure 1. Schematic diagram of the experimental setup.

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1. **QUESTION** ☐ **ANSWER** ☐

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1. A rectangular block of mass M and height h is placed on a horizontal surface. A force F is applied to the top edge of the block, directed at an angle θ to the horizontal. The block is on the verge of tipping over the right edge.

Find the magnitude of the force F in terms of M , h , and θ .

- (a) $F = \frac{Mh}{2} \tan \theta$
(b) $F = Mh \tan \theta$
(c) $F = \frac{Mh}{2} \cot \theta$
(d) $F = Mh \cot \theta$



2. A rectangular block of mass M and height h is placed on a horizontal surface. A force F is applied to the top edge of the block, directed at an angle θ to the horizontal. The block is on the verge of tipping over the right edge. Find the magnitude of the force F in terms of M , h , and θ .

- (a) $F = \frac{Mh}{2} \tan \theta$
(b) $F = Mh \tan \theta$
(c) $F = \frac{Mh}{2} \cot \theta$
(d) $F = Mh \cot \theta$

Ans: (a)
Ans: (b)



•

1. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top edge of the block, acting horizontally to the right. The block is on the verge of tipping over its right edge. Find the magnitude of the force F .

2. A uniform beam of length L and mass M is supported by a hinge at the left end and a cable at the right end. The cable is attached to the ceiling at a point directly above the hinge. The beam makes an angle θ with the horizontal. Find the tension in the cable.

- a) $F = \frac{Mgh}{2}$
- b) $F = \frac{Mgh}{4}$
- c) $F = \frac{Mgh}{8}$
- d) $F = \frac{Mgh}{16}$



- e) $F = \frac{Mgh}{32}$
- f) $F = \frac{Mgh}{64}$
- g) $F = \frac{Mgh}{128}$
- h) $F = \frac{Mgh}{256}$

3. A uniform beam of length L and mass M is supported by a hinge at the left end and a cable at the right end. The cable is attached to the ceiling at a point directly above the hinge. The beam makes an angle θ with the horizontal. Find the tension in the cable.

- a) $T = \frac{MgL}{2\cos\theta}$
- b) $T = \frac{MgL}{4\cos\theta}$
- c) $T = \frac{MgL}{8\cos\theta}$
- d) $T = \frac{MgL}{16\cos\theta}$

- e) $T = \frac{MgL}{32\cos\theta}$
- f) $T = \frac{MgL}{64\cos\theta}$
- g) $T = \frac{MgL}{128\cos\theta}$
- h) $T = \frac{MgL}{256\cos\theta}$



1. கீழ்க்கண்டவற்றைப் பற்றித் தகவல் கொடுக்கவும். (10 மதிப்புகள்)
 (அ) கீழ்க்கண்டவற்றைப் பற்றித் தகவல் கொடுக்கவும். (10 மதிப்புகள்)

கீழ்க்கண்டவற்றைப் பற்றித் தகவல் கொடுக்கவும்.

பெயர்	பெயர்	
பெயர்	பெயர்	
பெயர்	பெயர்	
பெயர்	பெயர்	

2. கீழ்க்கண்டவற்றைப் பற்றித் தகவல் கொடுக்கவும். (10 மதிப்புகள்)
 (அ) கீழ்க்கண்டவற்றைப் பற்றித் தகவல் கொடுக்கவும். (10 மதிப்புகள்)

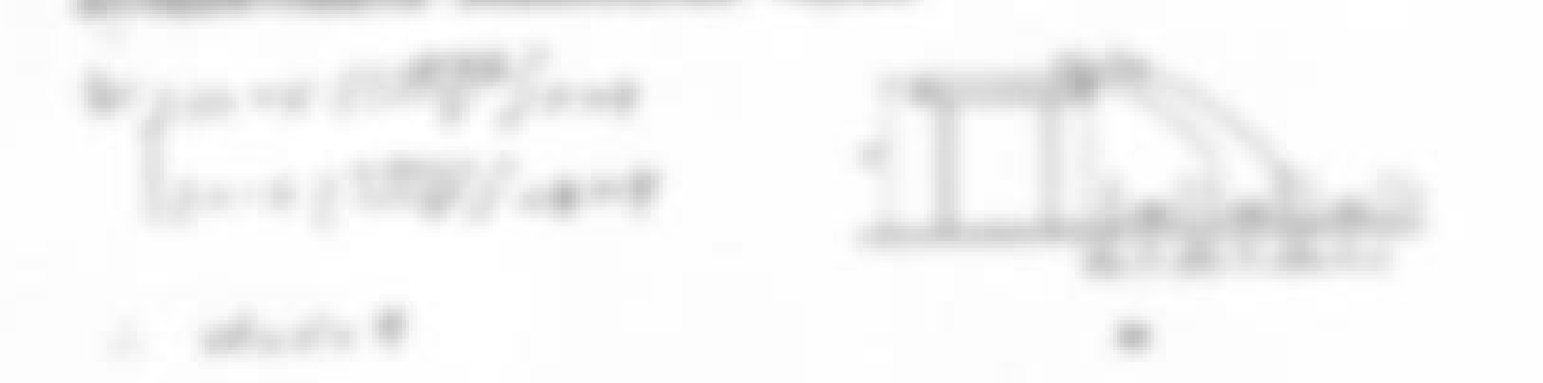
3. கீழ்க்கண்டவற்றைப் பற்றித் தகவல் கொடுக்கவும். (10 மதிப்புகள்)
 (அ) கீழ்க்கண்டவற்றைப் பற்றித் தகவல் கொடுக்கவும். (10 மதிப்புகள்)

① கீழ்க்கண்டவற்றைப் பற்றித் தகவல் கொடுக்கவும். ② கீழ்க்கண்டவற்றைப் பற்றித் தகவல் கொடுக்கவும்.	
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^a *Staphylococcus aureus* strains were isolated from the nasal cavity of patients with nasal polyps and from the nasal cavity of healthy subjects. *S. aureus* strains were isolated from the nasal cavity of patients with nasal polyps and from the nasal cavity of healthy subjects. *S. aureus* strains were isolated from the nasal cavity of patients with nasal polyps and from the nasal cavity of healthy subjects.



1. **QUESTION:** What is the purpose of the study?
2. **ANSWER:** The purpose of the study is to determine the effect of the intervention on the outcome.
3. **QUESTION:** What is the research design?
4. **ANSWER:** The research design is a randomized controlled trial.
5. **QUESTION:** What is the sample size?
6. **ANSWER:** The sample size is 1000 participants.
7. **QUESTION:** What is the primary outcome measure?
8. **ANSWER:** The primary outcome measure is the rate of adverse events.
9. **QUESTION:** What is the secondary outcome measure?
10. **ANSWER:** The secondary outcome measure is the rate of hospitalization.



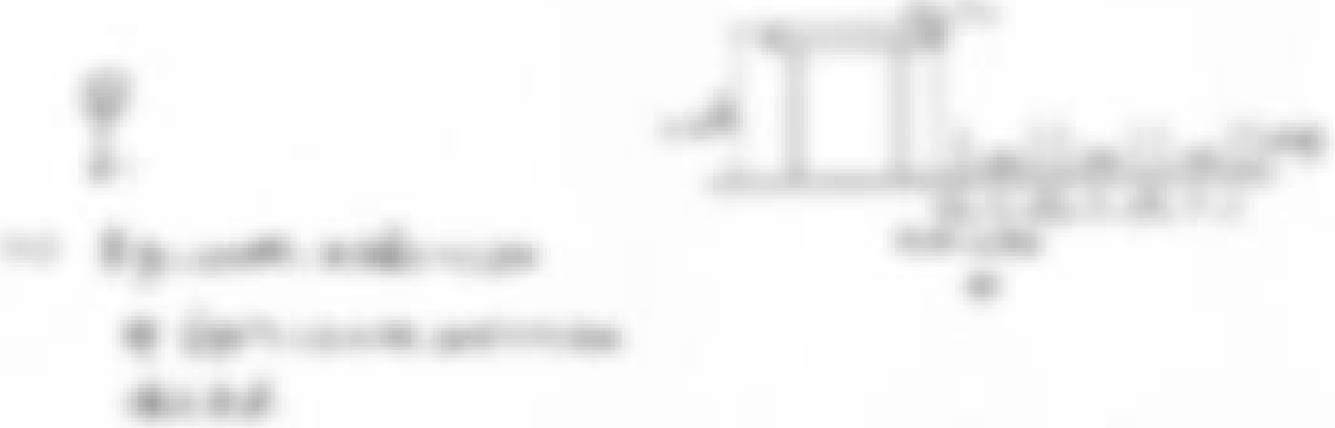
10. *Das erste Gespräch*

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1. **RESEARCH DESIGN**—This study was a descriptive, cross-sectional survey of the prevalence of self-reported depression among a representative sample of the adult population of the United States. The survey was conducted by the National Comorbidity Survey Replication (NCS-R), a large-scale, nationally representative survey of mental and physical disorders in the United States. The NCS-R is a follow-up of the National Comorbidity Survey (NCS), which was conducted in 1990-1992. The NCS-R was conducted in 2001-2002. The survey was conducted by the National Institute of Mental Health (NIMH) and the University of Michigan. The survey was conducted in two waves. The first wave was conducted in 2001 and the second wave was conducted in 2002. The survey was conducted in two waves. The first wave was conducted in 2001 and the second wave was conducted in 2002. The survey was conducted in two waves. The first wave was conducted in 2001 and the second wave was conducted in 2002.

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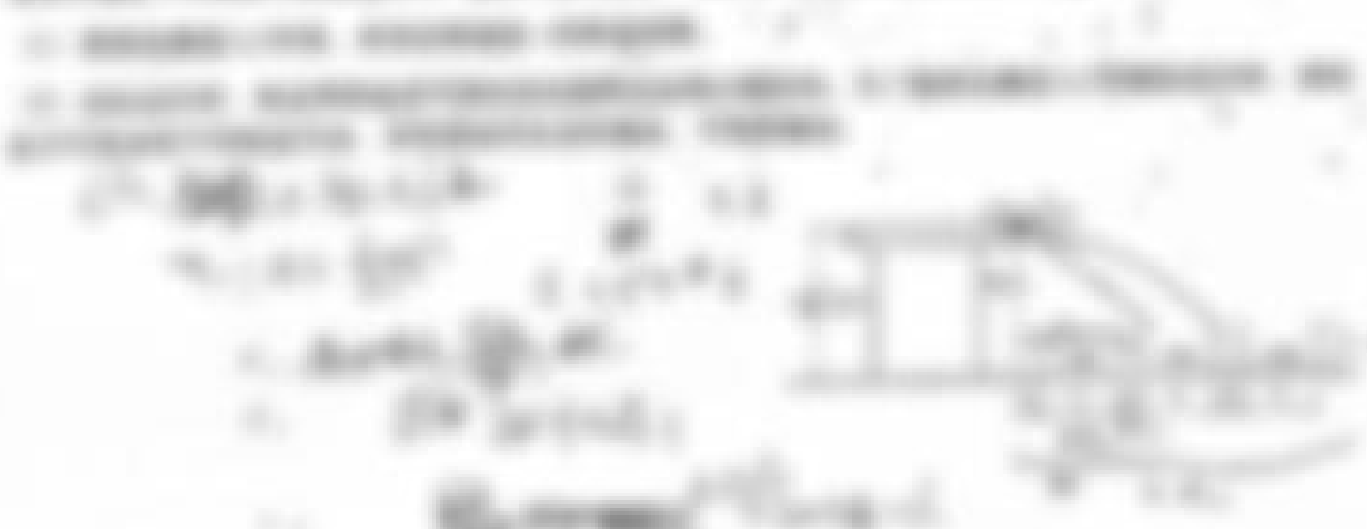
1. **QUESTION:** What is the purpose of the study?
2. **ANSWER:** The purpose of the study is to determine the effect of the intervention on the outcome.
3. **QUESTION:** What is the research design?
4. **ANSWER:** The research design is a randomized controlled trial.
5. **QUESTION:** What are the inclusion and exclusion criteria?
6. **ANSWER:** The inclusion criteria are: adults, aged 18 years and older, who are able to give informed consent. The exclusion criteria are: pregnant women, women with a history of heart disease, and women with a history of stroke.

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1. A rectangular block of material is shown in the diagram. The block is 10 cm long, 5 cm wide and 2 cm high. The mass of the block is 100 g. Calculate the density of the material.



2. A rectangular block of material is shown in the diagram. The block is 10 cm long, 5 cm wide and 2 cm high. The mass of the block is 100 g. Calculate the density of the material.



3. A rectangular block of material is shown in the diagram. The block is 10 cm long, 5 cm wide and 2 cm high. The mass of the block is 100 g. Calculate the density of the material.

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1. **Abstract:** This paper presents a new method for the automatic detection of the onset of a seizure. The method is based on the analysis of the non-linear properties of the EEG signal. The results show that the proposed method is able to detect the onset of a seizure with a high accuracy and a low false alarm rate.

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1. The following information is given for the company for the year ended 31st December 2018:

Particulars		Amount
Share capital		100,000
Reserves		20,000
Profit for the year		10,000
Dividend paid		5,000
Depreciation		2,000
Interest on loan		1,000
Income tax		1,000
Retained profit		5,000

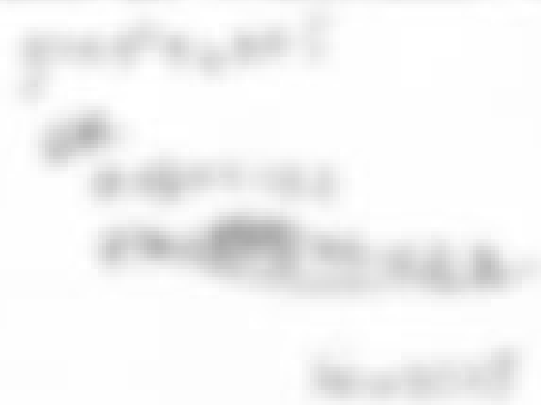
2. The company has decided to pay a dividend of 5% on the share capital for the year ended 31st December 2018.

3. The company has decided to pay a dividend of 5% on the share capital for the year ended 31st December 2018.



1. A rectangular block of material is shown in the figure. The block is subjected to a uniform pressure p acting on its top and bottom surfaces. The block is free to expand in the horizontal direction. The vertical displacement of the top surface is denoted by δ .

Consider the following two cases:



2. A rectangular block of material is shown in the figure. The block is subjected to a uniform pressure p acting on its top and bottom surfaces. The block is free to expand in the horizontal direction. The vertical displacement of the top surface is denoted by δ .

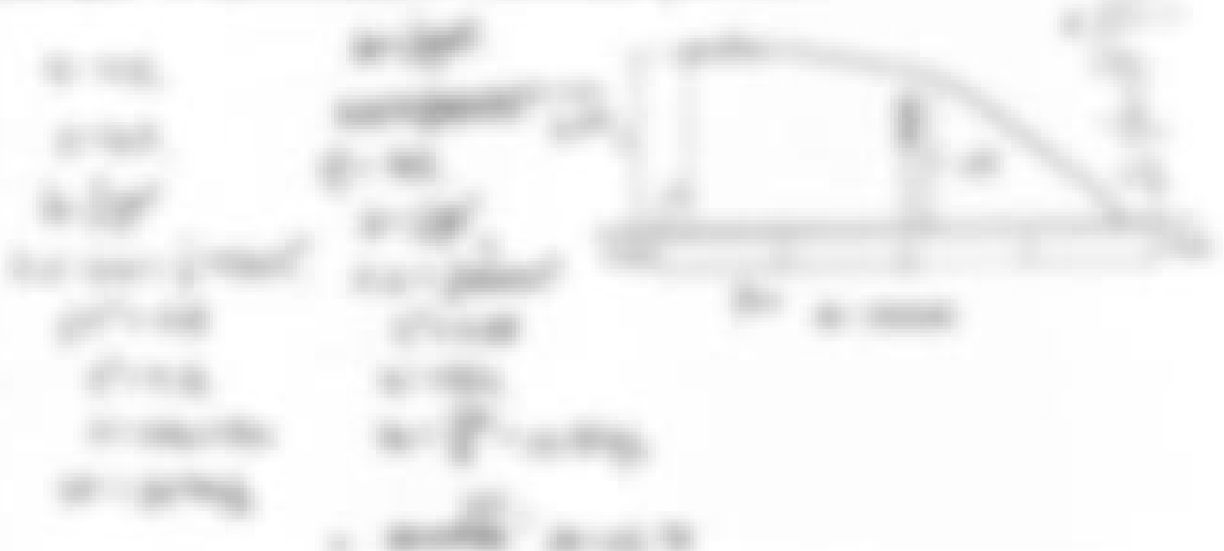
- (a) Consider the case where the block is subjected to a uniform pressure p acting on its top and bottom surfaces. The block is free to expand in the horizontal direction. The vertical displacement of the top surface is denoted by δ .
- (b) Consider the case where the block is subjected to a uniform pressure p acting on its top and bottom surfaces. The block is free to expand in the horizontal direction. The vertical displacement of the top surface is denoted by δ .



6/10/2020

1. A rectangular block of mass m is placed on a horizontal surface. A force F is applied to the block at an angle θ to the horizontal. The block is on the verge of moving. Find the coefficient of static friction μ_s between the block and the surface.

2. A rectangular block of mass m is placed on a horizontal surface. A force F is applied to the block at an angle θ to the horizontal. The block is on the verge of moving. Find the coefficient of static friction μ_s between the block and the surface.

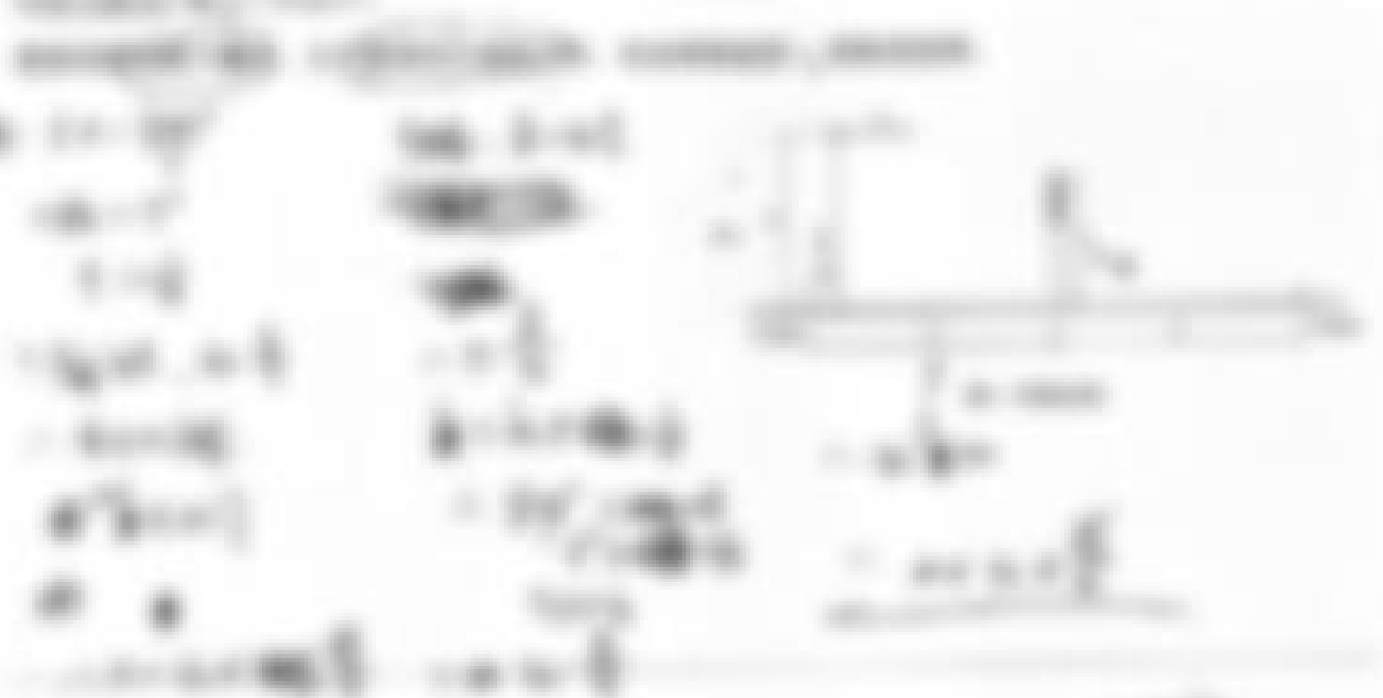


3. A rectangular block of mass m is placed on a horizontal surface. A force F is applied to the block at an angle θ to the horizontal. The block is on the verge of moving. Find the coefficient of static friction μ_s between the block and the surface.

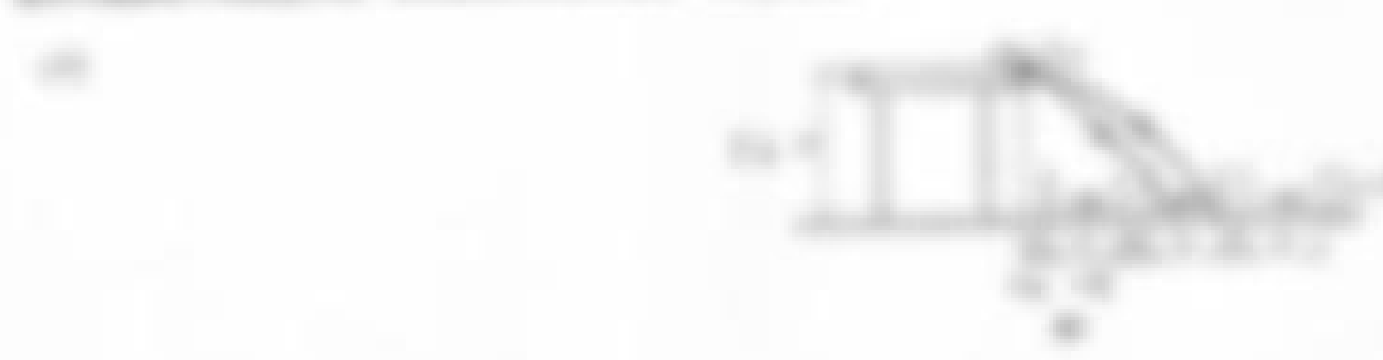
4. A rectangular block of mass m is placed on a horizontal surface. A force F is applied to the block at an angle θ to the horizontal. The block is on the verge of moving. Find the coefficient of static friction μ_s between the block and the surface.



1. The following information is given for the structure shown below. The structure is a continuous beam with three supports. The beam is divided into four segments by the supports. The segments are labeled 1, 2, 3, and 4. The supports are labeled A, B, and C. The beam is subjected to a uniformly distributed load of 10 kN/m over the entire length. The beam is also subjected to a point load of 20 kN at the midpoint of segment 2. The beam is supported by a pin support at A, a roller support at B, and a fixed support at C.



2. The following information is given for the structure shown below. The structure is a frame with two columns and one beam. The columns are labeled 1 and 2. The beam is labeled 3. The columns are subjected to a uniformly distributed load of 10 kN/m. The beam is subjected to a point load of 20 kN at the midpoint. The columns are supported by pin supports at the base. The beam is supported by a roller support at the right end.



3. The following information is given for the structure shown below. The structure is a frame with two columns and one beam. The columns are labeled 1 and 2. The beam is labeled 3. The columns are subjected to a uniformly distributed load of 10 kN/m. The beam is subjected to a point load of 20 kN at the midpoint. The columns are supported by pin supports at the base. The beam is supported by a roller support at the right end.



1. **QUESTION:** A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The top part is fixed to a wall, and the bottom part is free to move. The block is shown in the figure.

ANSWER: The block is shown in the figure.

The block is shown in the figure. The block is divided into two equal parts by a vertical plane. The top part is fixed to a wall, and the bottom part is free to move. The block is shown in the figure.



2. **QUESTION:** A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The top part is fixed to a wall, and the bottom part is free to move. The block is shown in the figure.

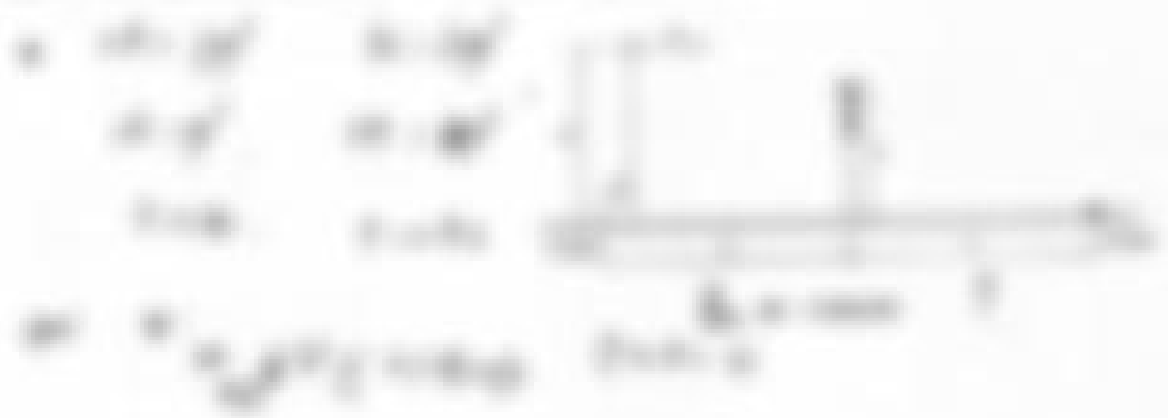
ANSWER: The block is shown in the figure.

3. **QUESTION:** A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The top part is fixed to a wall, and the bottom part is free to move. The block is shown in the figure.

The block is shown in the figure. The block is divided into two equal parts by a vertical plane. The top part is fixed to a wall, and the bottom part is free to move. The block is shown in the figure.



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1. **ප්‍රශ්නය:** පහත සඳහන් කරුණු සලකා බලා, ඒවායේ සත්‍ය බව හෝ අසත්‍ය බව තීරණය කරන්න.

ප්‍රකාශන: ප්‍රකාශන පහත දක්වා ඇත.

- 1. ප්‍රකාශන 1 සත්‍ය වේ.
- 2. ප්‍රකාශන 2 සත්‍ය වේ.
- 3. ප්‍රකාශන 3 සත්‍ය වේ.
- 4. ප්‍රකාශන 4 සත්‍ය වේ.
- 5. ප්‍රකාශන 5 සත්‍ය වේ.



ප්‍රකාශන 1 සත්‍ය වේ. ප්‍රකාශන 2 සත්‍ය වේ. ප්‍රකාශන 3 සත්‍ය වේ. ප්‍රකාශන 4 සත්‍ය වේ. ප්‍රකාශන 5 සත්‍ය වේ.

ප්‍රකාශන 1 සත්‍ය වේ.

2. **ප්‍රශ්නය:** පහත සඳහන් කරුණු සලකා බලා, ඒවායේ සත්‍ය බව හෝ අසත්‍ය බව තීරණය කරන්න.

ප්‍රකාශන: ප්‍රකාශන පහත දක්වා ඇත.

- 1. ප්‍රකාශන 1 සත්‍ය වේ.
- 2. ප්‍රකාශන 2 සත්‍ය වේ.
- 3. ප්‍රකාශන 3 සත්‍ය වේ.
- 4. ප්‍රකාශන 4 සත්‍ය වේ.
- 5. ප්‍රකාශන 5 සත්‍ය වේ.



ප්‍රකාශන 1 සත්‍ය වේ. ප්‍රකාශන 2 සත්‍ය වේ. ප්‍රකාශන 3 සත්‍ය වේ. ප්‍රකාශන 4 සත්‍ය වේ. ප්‍රකාශන 5 සත්‍ය වේ.

1. A horizontal beam of length 10 m is supported at its left end by a pin support and at its right end by a roller support. A uniformly distributed load of 2 kN/m is applied downwards over the entire length of the beam. Determine the reactions at the supports.

Solution: Consider the free-body diagram of the beam.

$$\sum F_y = 0$$

$$R_L + R_R - 20 = 0$$

$$R_L + R_R = 20$$

$$R_L = 20 - R_R$$

$$\sum M_A = 0$$

$$R_R(10) - 20(5) = 0$$

$$R_R = 10$$

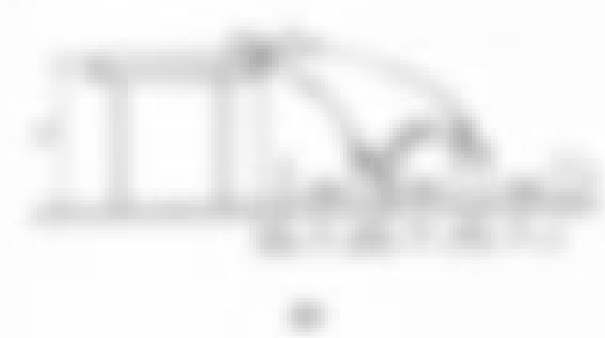


2. A horizontal beam of length 10 m is supported at its left end by a pin support and at its right end by a roller support. A point load of 10 kN is applied downwards at a distance of 3 m from the left end. Determine the reactions at the supports.

Solution: Consider the free-body diagram of the beam.

1. A horizontal beam of length 10 m is supported at its left end by a pin support and at its right end by a roller support. A point load of 10 kN is applied downwards at a distance of 3 m from the left end. Determine the reactions at the supports.

$$\sum F_y = 0$$



1. 某公司生产一种产品，其成本函数为 $C(x) = 0.01x^2 + 0.5x + 100$ ，其中 x 为产量。求当产量为 100 时的平均成本。

解：平均成本函数为 $\bar{C}(x) = \frac{C(x)}{x}$ 。

• $\bar{C}(100) = \frac{C(100)}{100}$

$$= \frac{0.01 \times 100^2 + 0.5 \times 100 + 100}{100}$$

• $\bar{C}(100) = 1.5$



2. 某公司生产一种产品，其成本函数为 $C(x) = 0.01x^2 + 0.5x + 100$ ，其中 x 为产量。求当产量为 100 时的平均成本。

解：平均成本函数为 $\bar{C}(x) = \frac{C(x)}{x}$ 。

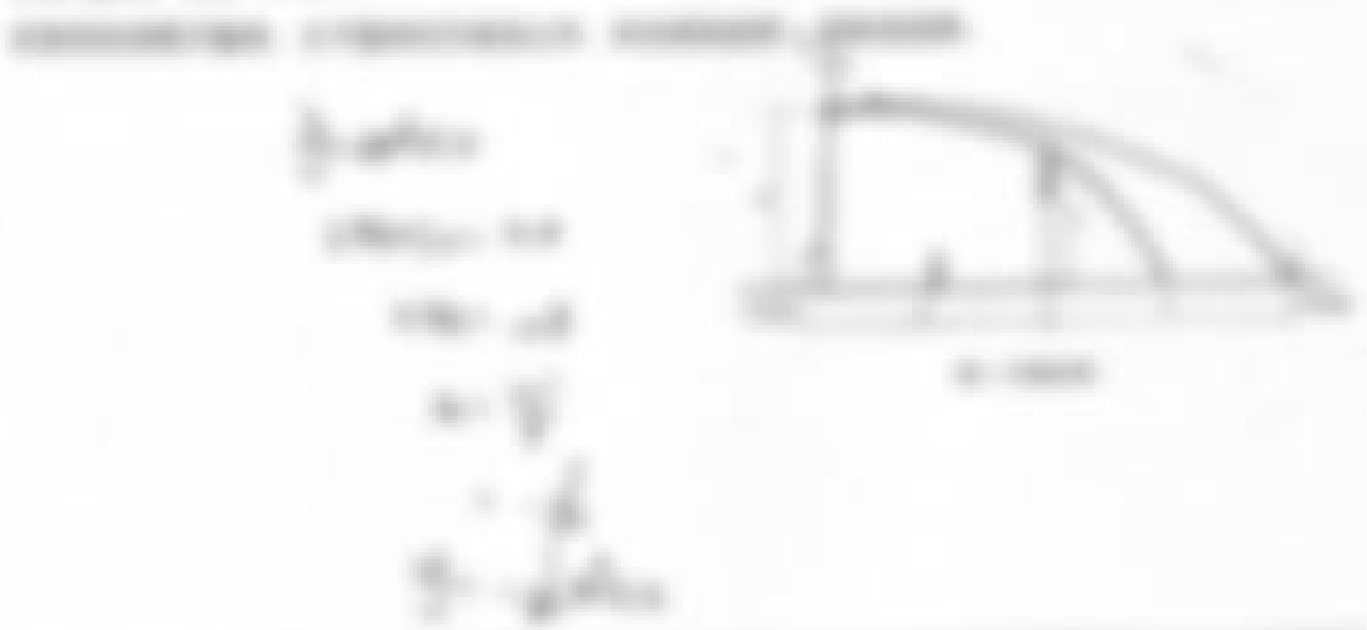
• $\bar{C}(100) = \frac{C(100)}{100}$

• $\bar{C}(100) = 1.5$

• $\bar{C}(100) = 1.5$

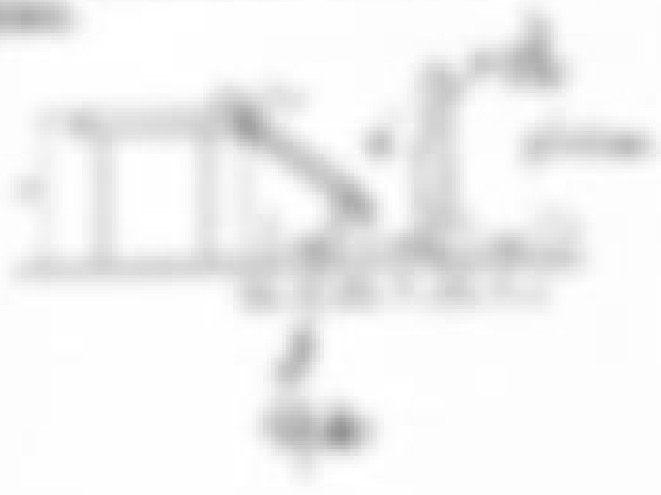


1. A particle of mass m is projected from the origin O of a Cartesian coordinate system with an initial velocity u at an angle α to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point P . The horizontal distance between O and P is R . The time taken for the particle to reach P is T . The maximum height reached by the particle is H . The range R is given by $R = \frac{u^2 \sin 2\alpha}{g}$. The time of flight T is given by $T = \frac{2u \sin \alpha}{g}$. The maximum height H is given by $H = \frac{u^2 \sin^2 \alpha}{2g}$.



2. A particle of mass m is projected from the origin O of a Cartesian coordinate system with an initial velocity u at an angle α to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point P . The horizontal distance between O and P is R . The time taken for the particle to reach P is T . The maximum height reached by the particle is H . The range R is given by $R = \frac{u^2 \sin 2\alpha}{g}$. The time of flight T is given by $T = \frac{2u \sin \alpha}{g}$. The maximum height H is given by $H = \frac{u^2 \sin^2 \alpha}{2g}$.

3. A particle of mass m is projected from the origin O of a Cartesian coordinate system with an initial velocity u at an angle α to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point P . The horizontal distance between O and P is R . The time taken for the particle to reach P is T . The maximum height reached by the particle is H . The range R is given by $R = \frac{u^2 \sin 2\alpha}{g}$. The time of flight T is given by $T = \frac{2u \sin \alpha}{g}$. The maximum height H is given by $H = \frac{u^2 \sin^2 \alpha}{2g}$.



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Journal of Internal Medicine 247: 395–402

1. **Abstract:** This paper presents a new method for the automatic detection of the onset of a seizure. The method is based on the analysis of the non-linear properties of the EEG signal. The results show that the proposed method is able to detect the onset of a seizure with a high accuracy and a low false alarm rate.

1. *Journal of the American Medical Association*, 1998; 279: 1001-1005.

1. *Explain the importance of the following factors in the development of a country's economy:*

1. A beam of length 10 m is supported at its left end by a pin support and at its right end by a roller support. A uniformly distributed load of 2 kN/m is applied over the entire length of the beam. Determine the reactions at the supports.

2. A beam of length 10 m is supported at its left end by a pin support and at its right end by a roller support. A point load of 10 kN is applied at the midpoint of the beam. Determine the reactions at the supports.



3. A beam of length 10 m is supported at its left end by a pin support and at its right end by a roller support. A point load of 10 kN is applied at the midpoint of the beam. Determine the reactions at the supports.

4. A beam of length 10 m is supported at its left end by a pin support and at its right end by a roller support. A point load of 10 kN is applied at the midpoint of the beam. Determine the reactions at the supports.

5. A beam of length 10 m is supported at its left end by a pin support and at its right end by a roller support. A point load of 10 kN is applied at the midpoint of the beam. Determine the reactions at the supports.

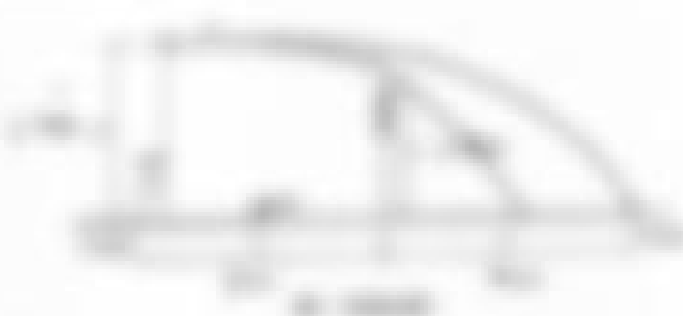
6. A beam of length 10 m is supported at its left end by a pin support and at its right end by a roller support. A point load of 10 kN is applied at the midpoint of the beam. Determine the reactions at the supports.



1. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined.

2. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined.

3. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined.



4. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined.

5. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined.

6. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined.



1. The following information was obtained from a survey of 100 students in a high school. The students were asked to indicate their favorite sport and whether they played it or not. The results are shown in the table below.

- 1. 100
- 2. 100
- 3. 100
- 4. 100
- 5. 100
- 6. 100
- 7. 100
- 8. 100
- 9. 100
- 10. 100



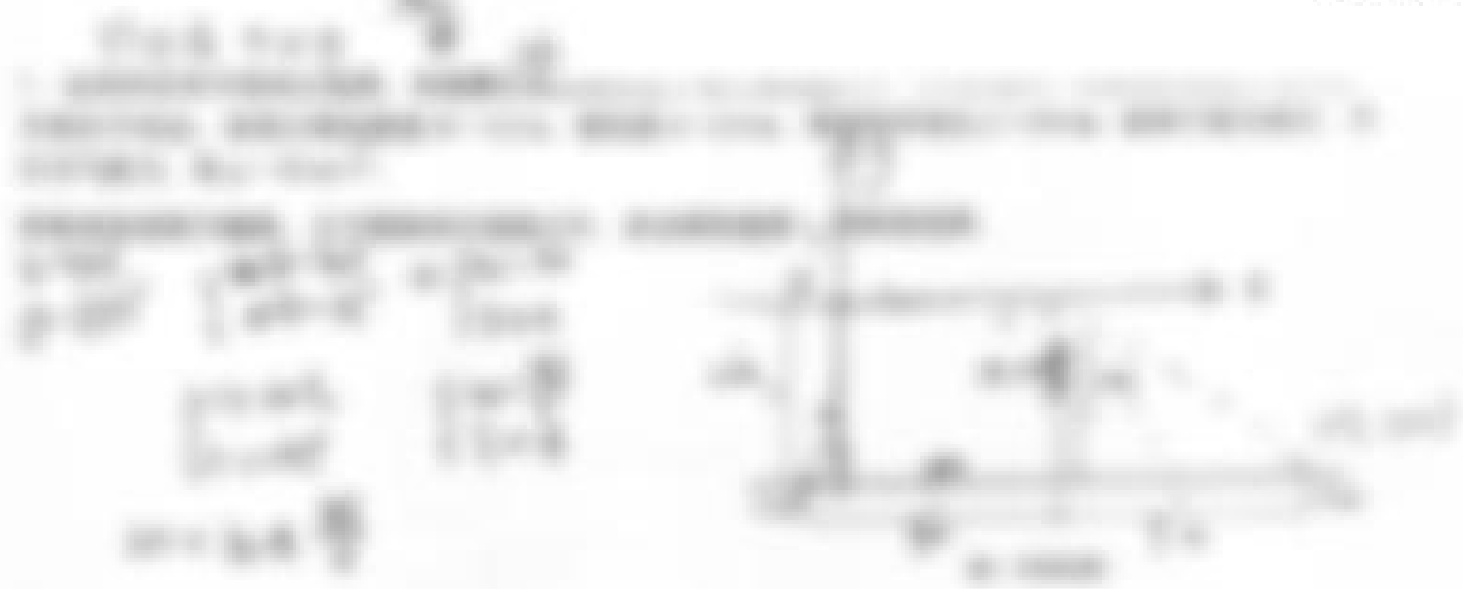
2. The following information was obtained from a survey of 100 students in a high school. The students were asked to indicate their favorite sport and whether they played it or not. The results are shown in the table below.

3. The following information was obtained from a survey of 100 students in a high school. The students were asked to indicate their favorite sport and whether they played it or not. The results are shown in the table below.

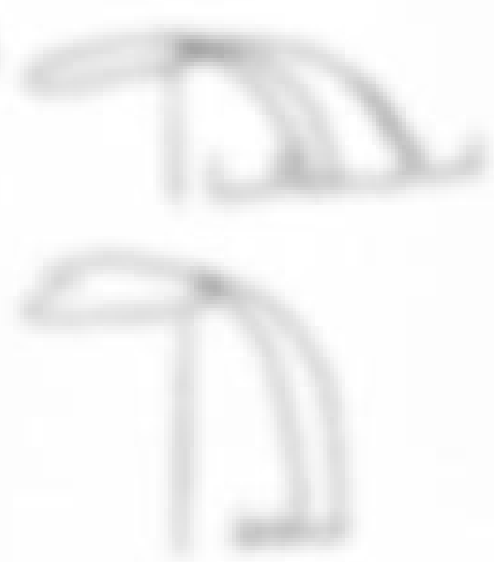
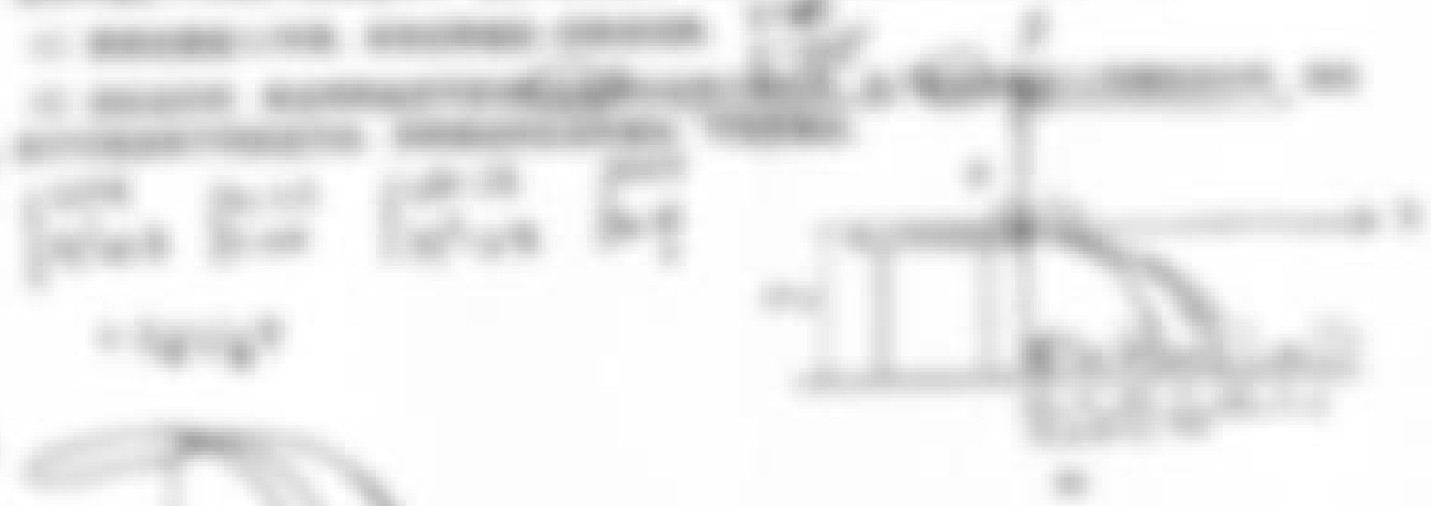
- 1. 100
- 2. 100
- 3. 100
- 4. 100
- 5. 100
- 6. 100
- 7. 100
- 8. 100
- 9. 100
- 10. 100



- 1. 100
- 2. 100
- 3. 100
- 4. 100
- 5. 100
- 6. 100
- 7. 100
- 8. 100
- 9. 100
- 10. 100



1. 本圖為...
 2. ...
 3. ...
 4. ...



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1. **RESEARCH DESIGN**—This study was a descriptive, cross-sectional study. The data were collected from a convenience sample of 100 nurses working in a tertiary care hospital in the city of Pune, India. The study was approved by the Institutional Review Board of the hospital. The data were collected from January to March 2018.

1. *Staphylococcus aureus* (Gram-positive, cocci in clusters)



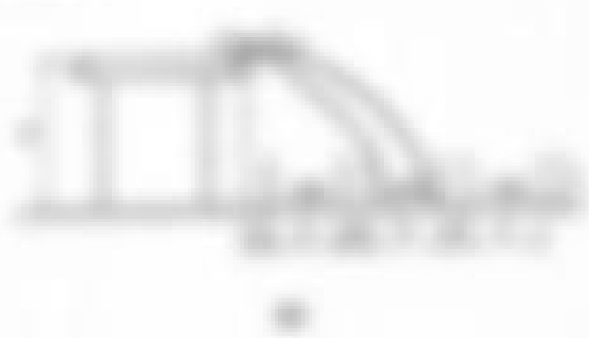
1. **Handwritten notes:**

- Handwritten text on the left side of the graph, possibly describing the function or its properties.

2. **Graph:**

3. **Annotations:**

- Labels 'a' and 'b' are present on the x-axis.
- A vertical line segment is drawn from the x-axis to the curve at point 'a'.
- A horizontal line segment is drawn from the y-axis to the curve at point 'b'.



1. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined.

2. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined.

- (a) $\sigma = 0$
- (b) $\sigma = \frac{1}{2} \sigma_0$
- (c) $\sigma = \sigma_0$
- (d) $\sigma = 2\sigma_0$



3. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined.

- (a) $\sigma = 0$
- (b) $\sigma = \frac{1}{2} \sigma_0$
- (c) $\sigma = \sigma_0$
- (d) $\sigma = 2\sigma_0$

4. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined.



1. **QUESTION:** A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined from the equilibrium of the two parts.

ANSWER: The stress σ is to be determined from the equilibrium of the two parts.

$$\begin{aligned} \sigma &= \frac{F}{A} \\ \sigma &= \frac{P}{A} \\ \sigma &= \frac{W}{A} \\ \sigma &= \frac{M}{A} \\ \sigma &= \frac{V}{A} \end{aligned}$$



$$\begin{aligned} \sigma &= \frac{F}{A} \\ \sigma &= \frac{P}{A} \\ \sigma &= \frac{W}{A} \\ \sigma &= \frac{M}{A} \\ \sigma &= \frac{V}{A} \end{aligned}$$

2. **QUESTION:** A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined from the equilibrium of the two parts.

ANSWER: The stress σ is to be determined from the equilibrium of the two parts.

3. **QUESTION:** A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The horizontal faces are free of stress. The block is divided into two equal parts by a vertical plane. The stress σ is to be determined from the equilibrium of the two parts.



1. Geometrische Konstruktion (Zirkel und Lineal)
2. Algebra (Gleichungen, Ungleichungen)
3. Trigonometrie (Sinus, Cosinus, Tangens)
4. Komplexe Zahlen (Euler'sche Formel)

Geometrische Konstruktion

1. Grundkonstruktionen

- a) Strecke übertragen
- b) Winkel übertragen
- c) Senkrechte und Parallele

2. Geometrische Konstruktion

3. Algebra

4. Trigonometrie

5. Komplexe Zahlen



1. Geometrische Konstruktion (Zirkel und Lineal)
2. Algebra (Gleichungen, Ungleichungen)
3. Trigonometrie (Sinus, Cosinus, Tangens)
4. Komplexe Zahlen (Euler'sche Formel)

Geometrische Konstruktion

1. Grundkonstruktionen

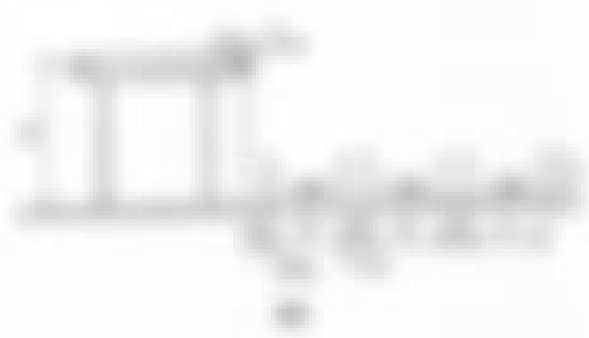
- a) Strecke übertragen
- b) Winkel übertragen
- c) Senkrechte und Parallele

2. Geometrische Konstruktion

3. Algebra

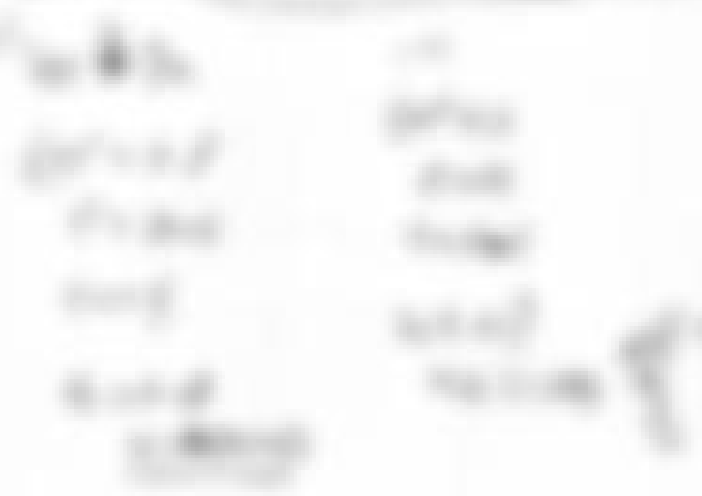
4. Trigonometrie

5. Komplexe Zahlen



1. A rectangular plate of width b and height h is subjected to a uniformly distributed load w acting downwards. The plate is supported by a pin support at the bottom left corner and a roller support at the bottom right corner. The plate is divided into two equal parts by a vertical line passing through the center. The left part is labeled 'Left part' and the right part is labeled 'Right part'.

2. A rectangular plate of width b and height h is subjected to a uniformly distributed load w acting downwards. The plate is supported by a pin support at the bottom left corner and a roller support at the bottom right corner. The plate is divided into two equal parts by a vertical line passing through the center. The left part is labeled 'Left part' and the right part is labeled 'Right part'.



3. A rectangular plate of width b and height h is subjected to a uniformly distributed load w acting downwards. The plate is supported by a pin support at the bottom left corner and a roller support at the bottom right corner. The plate is divided into two equal parts by a vertical line passing through the center. The left part is labeled 'Left part' and the right part is labeled 'Right part'.

1. A rectangular plate of width b and height h is subjected to a uniformly distributed load w acting downwards. The plate is supported by a pin support at the bottom left corner and a roller support at the bottom right corner. The plate is divided into two equal parts by a vertical line passing through the center. The left part is labeled 'Left part' and the right part is labeled 'Right part'.

2. A rectangular plate of width b and height h is subjected to a uniformly distributed load w acting downwards. The plate is supported by a pin support at the bottom left corner and a roller support at the bottom right corner. The plate is divided into two equal parts by a vertical line passing through the center. The left part is labeled 'Left part' and the right part is labeled 'Right part'.



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1. *Abstracts* (1990-1991) (1991) 100-101. 2. *Abstracts* (1992-1993) (1993) 100-101. 3. *Abstracts* (1994-1995) (1995) 100-101. 4. *Abstracts* (1996-1997) (1997) 100-101. 5. *Abstracts* (1998-1999) (1999) 100-101. 6. *Abstracts* (2000-2001) (2001) 100-101. 7. *Abstracts* (2002-2003) (2003) 100-101. 8. *Abstracts* (2004-2005) (2005) 100-101. 9. *Abstracts* (2006-2007) (2007) 100-101. 10. *Abstracts* (2008-2009) (2009) 100-101. 11. *Abstracts* (2010-2011) (2011) 100-101. 12. *Abstracts* (2012-2013) (2013) 100-101. 13. *Abstracts* (2014-2015) (2015) 100-101. 14. *Abstracts* (2016-2017) (2017) 100-101. 15. *Abstracts* (2018-2019) (2019) 100-101. 16. *Abstracts* (2020-2021) (2021) 100-101. 17. *Abstracts* (2022-2023) (2023) 100-101. 18. *Abstracts* (2024-2025) (2025) 100-101. 19. *Abstracts* (2026-2027) (2027) 100-101. 20. *Abstracts* (2028-2029) (2029) 100-101. 21. *Abstracts* (2030-2031) (2031) 100-101. 22. *Abstracts* (2032-2033) (2033) 100-101. 23. *Abstracts* (2034-2035) (2035) 100-101. 24. *Abstracts* (2036-2037) (2037) 100-101. 25. *Abstracts* (2038-2039) (2039) 100-101. 26. *Abstracts* (2040-2041) (2041) 100-101. 27. *Abstracts* (2042-2043) (2043) 100-101. 28. *Abstracts* (2044-2045) (2045) 100-101. 29. *Abstracts* (2046-2047) (2047) 100-101. 30. *Abstracts* (2048-2049) (2049) 100-101. 31. *Abstracts* (2050-2051) (2051) 100-101. 32. *Abstracts* (2052-2053) (2053) 100-101. 33. *Abstracts* (2054-2055) (2055) 100-101. 34. *Abstracts* (2056-2057) (2057) 100-101. 35. *Abstracts* (2058-2059) (2059) 100-101. 36. *Abstracts* (2060-2061) (2061) 100-101. 37. *Abstracts* (2062-2063) (2063) 100-101. 38. *Abstracts* (2064-2065) (2065) 100-101. 39. *Abstracts* (2066-2067) (2067) 100-101. 40. *Abstracts* (2068-2069) (2069) 100-101. 41. *Abstracts* (2070-2071) (2071) 100-101. 42. *Abstracts* (2072-2073) (2073) 100-101. 43. *Abstracts* (2074-2075) (2075) 100-101. 44. *Abstracts* (2076-2077) (2077) 100-101. 45. *Abstracts* (2078-2079) (2079) 100-101. 46. *Abstracts* (2080-2081) (2081) 100-101. 47. *Abstracts* (2082-2083) (2083) 100-101. 48. *Abstracts* (2084-2085) (2085) 100-101. 49. *Abstracts* (2086-2087) (2087) 100-101. 50. *Abstracts* (2088-2089) (2089) 100-101. 51. *Abstracts* (2090-2091) (2091) 100-101. 52. *Abstracts* (2092-2093) (2093) 100-101. 53. *Abstracts* (2094-2095) (2095) 100-101. 54. *Abstracts* (2096-2097) (2097) 100-101. 55. *Abstracts* (2098-2099) (2099) 100-101. 56. *Abstracts* (2100-2101) (2101) 100-101. 57. *Abstracts* (2102-2103) (2103) 100-101. 58. *Abstracts* (2104-2105) (2105) 100-101. 59. *Abstracts* (2106-2107) (2107) 100-101. 60. *Abstracts* (2108-2109) (2109) 100-101. 61. *Abstracts* (2110-2111) (2111) 100-101. 62. *Abstracts* (2112-2113) (2113) 100-101. 63. *Abstracts* (2114-2115) (2115) 100-101. 64. *Abstracts* (2116-2117) (2117) 100-101. 65. *Abstracts* (2118-2119) (2119) 100-101. 66. *Abstracts* (2120-2121) (2121) 100-101. 67. *Abstracts* (2122-2123) (2123) 100-101. 68. *Abstracts* (2124-2125) (2125) 100-101. 69. *Abstracts* (2126-2127) (2127) 100-101. 70. *Abstracts* (2128-2129) (2129) 100-101. 71. *Abstracts* (2130-2131) (2131) 100-101. 72. *Abstracts* (2132-2133) (2133) 100-101. 73. *Abstracts* (2134-2135) (2135) 100-101. 74. *Abstracts* (2136-2137) (2137) 100-101. 75. *Abstracts* (2138-2139) (2139) 100-101. 76. *Abstracts* (2140-2141) (2141) 100-101. 77. *Abstracts* (2142-2143) (2143) 100-101. 78. *Abstracts* (2144-2145) (2145) 100-101. 79. *Abstracts* (2146-2147) (2147) 100-101. 80. *Abstracts* (2148-2149) (2149) 100-101. 81. *Abstracts* (2150-2151) (2151) 100-101. 82. *Abstracts* (2152-2153) (2153) 100-101. 83. *Abstracts* (2154-2155) (2155) 100-101. 84. *Abstracts* (2156-2157) (2157) 100-101. 85. *Abstracts* (2158-2159) (2159) 100-101. 86. *Abstracts* (2160-2161) (2161) 100-101. 87. *Abstracts* (2162-2163) (2163) 100-101. 88. *Abstracts* (2164-2165) (2165) 100-101. 89. *Abstracts* (2166-2167) (2167) 100-101. 90. *Abstracts* (2168-2169) (2169) 100-101. 91. *Abstracts* (2170-2171) (2171) 100-101. 92. *Abstracts* (2172-2173) (2173) 100-101. 93. *Abstracts* (2174-2175) (2175) 100-101. 94. *Abstracts* (2176-2177) (2177) 100-101. 95. *Abstracts* (2178-2179) (2179) 100-101. 96. *Abstracts* (2180-2181) (2181) 100-101. 97. *Abstracts* (2182-2183) (2183) 100-101. 98. *Abstracts* (2184-2185) (2185) 100-101. 99. *Abstracts* (2186-2187) (2187) 100-101. 100. *Abstracts* (2188-2189) (2189) 100-101. 101. *Abstracts* (2190-2191) (2191) 100-101. 102. *Abstracts* (2192-2193) (2193) 100-101. 103. *Abstracts* (2194-2195) (2195) 100-101. 104. *Abstracts* (2196-2197) (2197) 100-101. 105. *Abstracts* (2198-2199) (2199) 100-101. 106. *Abstracts* (2200-2201) (2201) 100-101. 107. *Abstracts* (2202-2203) (2203) 100-101. 108. *Abstracts* (2204-2205) (2205)

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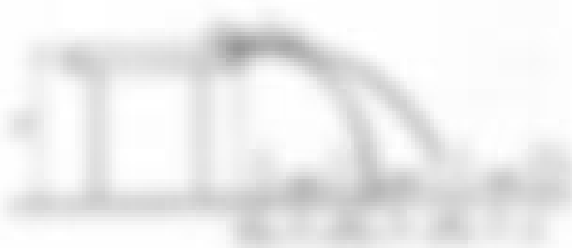
1. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top right corner of the block, acting horizontally to the right. The block is on the verge of tipping over the right edge. Find the magnitude of the force F .

2. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top right corner of the block, acting horizontally to the right. The block is on the verge of tipping over the right edge. Find the magnitude of the force F .



3. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top right corner of the block, acting horizontally to the right. The block is on the verge of tipping over the right edge. Find the magnitude of the force F .

4. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top right corner of the block, acting horizontally to the right. The block is on the verge of tipping over the right edge. Find the magnitude of the force F .



1. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top right corner of the block, directed at an angle θ to the horizontal. The block is on the verge of tipping clockwise about its bottom-left corner. Determine the magnitude of the force F in terms of m , h , and θ .

2. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top right corner of the block, directed at an angle θ to the horizontal. The block is on the verge of tipping clockwise about its bottom-left corner. Determine the magnitude of the force F in terms of m , h , and θ .

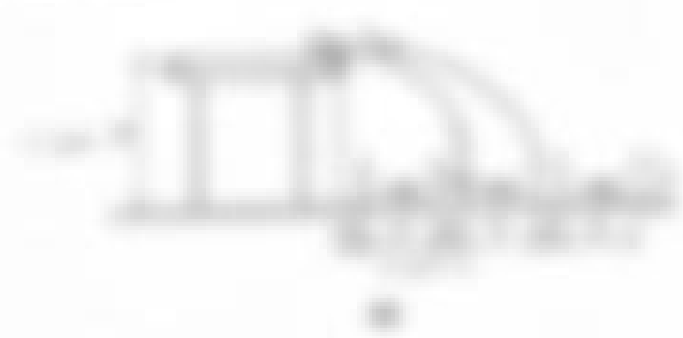
- (a) $F = \frac{mgh}{2 \sin \theta}$
- (b) $F = \frac{mgh}{2 \cos \theta}$
- (c) $F = \frac{mgh}{2 \tan \theta}$
- (d) $F = \frac{mgh}{2 \cot \theta}$



3. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top right corner of the block, directed at an angle θ to the horizontal. The block is on the verge of tipping clockwise about its bottom-left corner. Determine the magnitude of the force F in terms of m , h , and θ .

4. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top right corner of the block, directed at an angle θ to the horizontal. The block is on the verge of tipping clockwise about its bottom-left corner. Determine the magnitude of the force F in terms of m , h , and θ .

5. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top right corner of the block, directed at an angle θ to the horizontal. The block is on the verge of tipping clockwise about its bottom-left corner. Determine the magnitude of the force F in terms of m , h , and θ .



- (a) $F = \frac{mgh}{2 \sin \theta}$
- (b) $F = \frac{mgh}{2 \cos \theta}$
- (c) $F = \frac{mgh}{2 \tan \theta}$
- (d) $F = \frac{mgh}{2 \cot \theta}$

Figure 3

1. The following information is given for the structure shown in Figure 3. The structure is a frame with a horizontal beam of length 10 m and a vertical column of height 4 m. The beam is supported by a pin support at the left end and a roller support at the right end. The column is supported by a pin support at the base. A uniformly distributed load of 2 kN/m is applied vertically downwards over the entire length of the beam. A point load of 10 kN is applied vertically downwards at the free end of the beam (the right end).

2. Determine the reactions at the supports and the internal forces in the beam and column.

- (a) Reactions at the supports
- (b) Internal forces in the beam
- (c) Internal forces in the column
- (d) Shear force diagram
- (e) Bending moment diagram
- (f) Axial force diagram



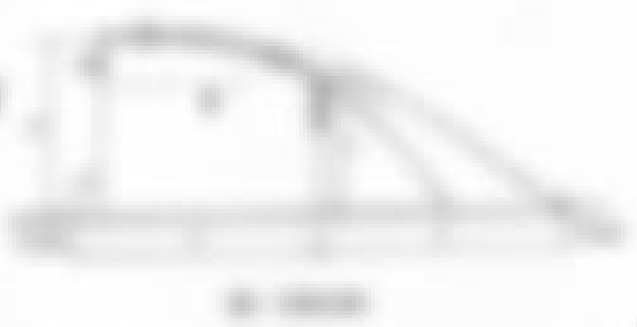
3. The following information is given for the structure shown in Figure 4. The structure is a frame with a horizontal beam of length 10 m and a vertical column of height 4 m. The beam is supported by a pin support at the left end and a roller support at the right end. The column is supported by a pin support at the base. A uniformly distributed load of 2 kN/m is applied vertically downwards over the entire length of the beam. A point load of 10 kN is applied vertically downwards at the free end of the beam (the right end).

- (a) Reactions at the supports
- (b) Internal forces in the beam
- (c) Internal forces in the column
- (d) Shear force diagram
- (e) Bending moment diagram
- (f) Axial force diagram



1. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The stress σ is acting on the vertical faces of the block.

2. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The stress σ is acting on the vertical faces of the block.



3. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The stress σ is acting on the vertical faces of the block.

4. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The stress σ is acting on the vertical faces of the block.

5. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The stress σ is acting on the vertical faces of the block.

6. A rectangular block of material is shown in the figure. The block is subjected to a uniform stress σ acting on the vertical faces. The block is divided into two equal parts by a vertical plane. The stress σ is acting on the vertical faces of the block.



1. **ଉପରୋକ୍ତ ଉପାଦାନଗୁଡ଼ିକରୁ ଉପଯୁକ୍ତ ଉପାଦାନଗୁଡ଼ିକ ଚିହ୍ନଟ କରାଯାଇ, ଉପଯୁକ୍ତ ଉପାଦାନଗୁଡ଼ିକର ନାମ ଓ ଉପାଦାନଗୁଡ଼ିକର ଉପାଦାନ ଗୁଣାଯିବ।**

ଉପରୋକ୍ତ ଉପାଦାନଗୁଡ଼ିକର ନାମ ଓ ଉପାଦାନ ଗୁଣାଯିବ।

ପ୍ରଥମ ଉପାଦାନ ଗୁଣାଯିବ।
ଉପାଦାନ ଗୁଣାଯିବ।



ପ୍ରଥମ ଉପାଦାନ ଗୁଣାଯିବ।
ଉପାଦାନ ଗୁଣାଯିବ।
ଉପାଦାନ ଗୁଣାଯିବ।
ଉପାଦାନ ଗୁଣାଯିବ।



2. **ଉପରୋକ୍ତ ଉପାଦାନଗୁଡ଼ିକରୁ ଉପଯୁକ୍ତ ଉପାଦାନଗୁଡ଼ିକ ଚିହ୍ନଟ କରାଯାଇ, ଉପଯୁକ୍ତ ଉପାଦାନଗୁଡ଼ିକର ନାମ ଓ ଉପାଦାନଗୁଡ଼ିକର ଉପାଦାନ ଗୁଣାଯିବ।**

3. **ଉପରୋକ୍ତ ଉପାଦାନଗୁଡ଼ିକରୁ ଉପଯୁକ୍ତ ଉପାଦାନଗୁଡ଼ିକ ଚିହ୍ନଟ କରାଯାଇ, ଉପଯୁକ୍ତ ଉପାଦାନଗୁଡ଼ିକର ନାମ ଓ ଉପାଦାନଗୁଡ଼ିକର ଉପାଦାନ ଗୁଣାଯିବ।**

ଉପାଦାନ ଗୁଣାଯିବ।
ଉପାଦାନ ଗୁଣାଯିବ।
ଉପାଦାନ ଗୁଣାଯିବ।
ଉପାଦାନ ଗୁଣାଯିବ।
ଉପାଦାନ ଗୁଣାଯିବ।



ଉପାଦାନ ଗୁଣାଯିବ।
ଉପାଦାନ ଗୁଣାଯିବ।
ଉପାଦାନ ଗୁଣାଯିବ।

7-5-11 9

1. The following information was obtained from a survey of the population of a certain city. The population is divided into four groups, A, B, C, and D, and the following information was obtained:

Group A: 1000, Group B: 2000, Group C: 3000, Group D: 4000

Age	10-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100
Group A	100	150	200	250	300	350	400	450	500
Group B	150	200	250	300	350	400	450	500	550
Group C	200	250	300	350	400	450	500	550	600
Group D	250	300	350	400	450	500	550	600	650



2. The following information was obtained from a survey of the population of a certain city. The population is divided into four groups, A, B, C, and D, and the following information was obtained:

Group A: 1000, Group B: 2000, Group C: 3000, Group D: 4000

Age	10-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100
Group A	100	150	200	250	300	350	400	450	500
Group B	150	200	250	300	350	400	450	500	550
Group C	200	250	300	350	400	450	500	550	600
Group D	250	300	350	400	450	500	550	600	650



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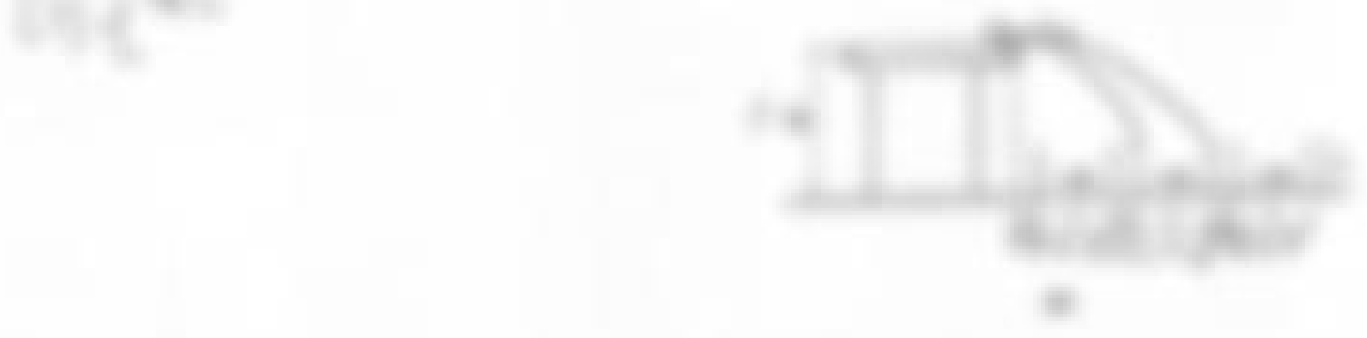
1. A rectangular block of mass m is placed on a horizontal surface. A force F is applied to the block at an angle θ to the horizontal. The block is on the verge of moving. Find the coefficient of friction μ between the block and the surface.

2. A rectangular block of mass m is placed on a horizontal surface. A force F is applied to the block at an angle θ to the horizontal. The block is on the verge of moving. Find the coefficient of friction μ between the block and the surface.



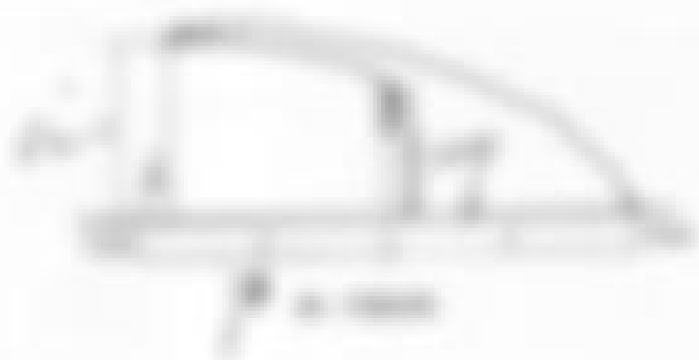
3. A rectangular block of mass m is placed on a horizontal surface. A force F is applied to the block at an angle θ to the horizontal. The block is on the verge of moving. Find the coefficient of friction μ between the block and the surface.

4. A rectangular block of mass m is placed on a horizontal surface. A force F is applied to the block at an angle θ to the horizontal. The block is on the verge of moving. Find the coefficient of friction μ between the block and the surface.



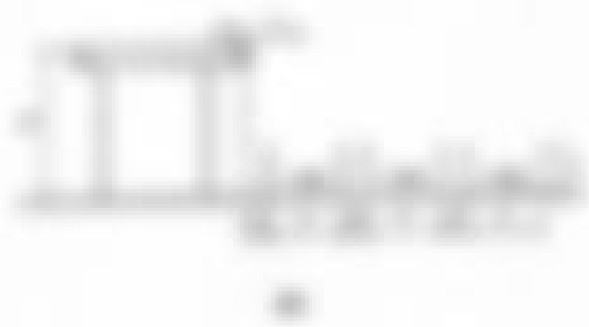
1. A rectangular block of material is shown in the figure. The block is subjected to a uniform pressure p acting on its top and bottom surfaces. The block is free to expand in the horizontal direction. The vertical displacement of the top surface is denoted by δ .

2. The block is made of a material with a Young's modulus E and a Poisson's ratio ν .

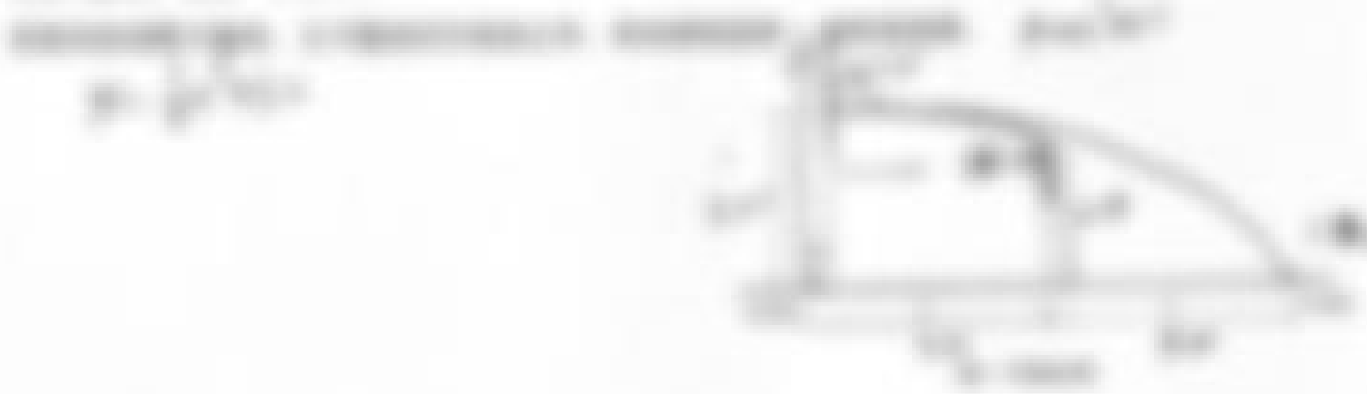


3. The block is supported by a rigid base. The base is fixed and does not move. The block is free to expand in the horizontal direction. The vertical displacement of the top surface is denoted by δ .

4. The block is made of a material with a Young's modulus E and a Poisson's ratio ν . The block is free to expand in the horizontal direction. The vertical displacement of the top surface is denoted by δ .



1. The following information is given for the structure shown below. The structure is a frame with a horizontal beam of length 10 m and a vertical column of height 4 m. The beam is supported by a pin support at the left end and a roller support at the right end. The column is supported by a pin support at the base. A uniformly distributed load of 2 kN/m is applied vertically downwards over the entire length of the beam. A point load of 10 kN is applied vertically downwards at the free end of the beam. The angle between the beam and the column is 45 degrees.



2. The following information is given for the structure shown below. The structure is a frame with a horizontal beam of length 10 m and a vertical column of height 4 m. The beam is supported by a pin support at the left end and a roller support at the right end. The column is supported by a pin support at the base. A uniformly distributed load of 2 kN/m is applied vertically downwards over the entire length of the beam. A point load of 10 kN is applied vertically downwards at the free end of the beam. The angle between the beam and the column is 45 degrees.

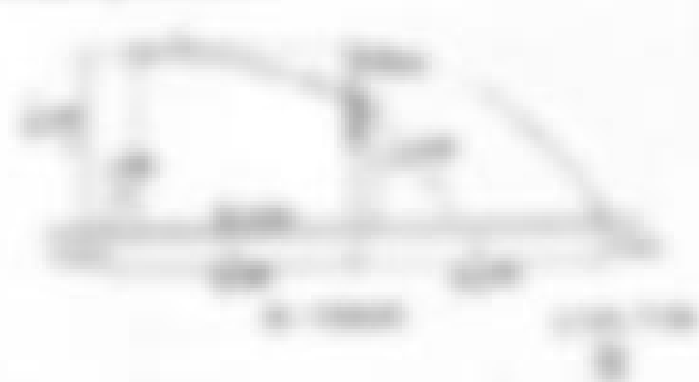
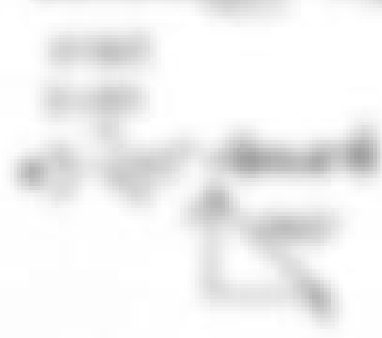
- a. Determine the reaction forces.
- b. Draw the shear force diagram and bending moment diagram.





1. The following information is given for the reaction of hydrogen gas with chlorine gas to form hydrogen chloride gas. The reaction is exothermic.

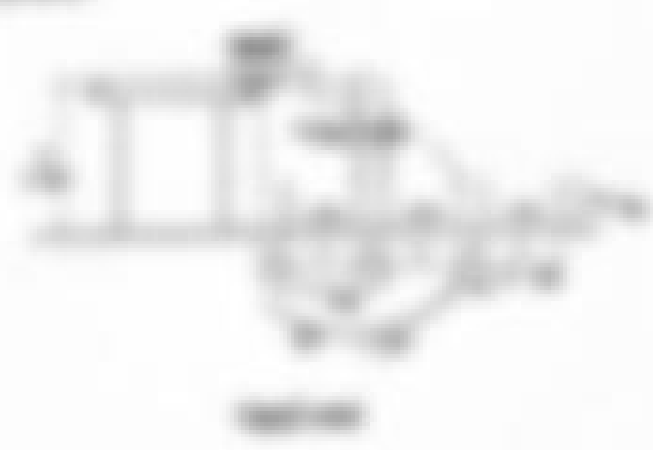
$$\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g}) \quad \Delta H^\circ_{\text{rxn}} = -184.6 \text{ kJ/mol}$$



2. The following information is given for the reaction of hydrogen gas with chlorine gas to form hydrogen chloride gas. The reaction is exothermic.

$$\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{HCl}(\text{g}) \quad \Delta H^\circ_{\text{rxn}} = -184.6 \text{ kJ/mol}$$

- a. Calculate the standard enthalpy of formation of hydrogen chloride gas.
- b. Calculate the standard enthalpy of formation of hydrogen gas.



1. **Introduction** - This document is a **technical specification** for the **development** of a **new product**. It is intended to provide a **clear and concise** description of the **requirements** for the **product** and to serve as a **reference** for the **development team**.

2. **Scope** - The scope of this document is limited to the **development** of the **product** and does not include the **design** or **manufacture** of the **product**.

3. **References** - The following references are used in this document:

- 1. **Technical Specification** - This document is a **technical specification** for the **development** of a **new product**.
- 2. **Design** - This document is a **design** for the **product**.
- 3. **Manufacture** - This document is a **manufacture** for the **product**.

4. **Definitions** - The following definitions are used in this document:

- 1. **Product** - A **product** is a **product** that is **developed** for the **purpose** of **meeting** the **requirements** of the **customer**.
- 2. **Design** - A **design** is a **design** that is **developed** for the **purpose** of **meeting** the **requirements** of the **customer**.
- 3. **Manufacture** - A **manufacture** is a **manufacture** that is **developed** for the **purpose** of **meeting** the **requirements** of the **customer**.

5. **Requirements** - The requirements for the **product** are as follows:

- 1. **Performance** - The **product** must be able to **perform** at a **level** of **performance** that is **consistent** with the **requirements** of the **customer**.
- 2. **Reliability** - The **product** must be able to **reliably** **perform** at a **level** of **performance** that is **consistent** with the **requirements** of the **customer**.
- 3. **Cost** - The **product** must be able to **perform** at a **level** of **performance** that is **consistent** with the **requirements** of the **customer**.

6. **Conclusion** - This document is a **technical specification** for the **development** of a **new product**. It is intended to provide a **clear and concise** description of the **requirements** for the **product** and to serve as a **reference** for the **development team**.

1. **Introduction** - This document is a **technical specification** for the **development** of a **new product**. It is intended to provide a **clear and concise** description of the **requirements** for the **product** and to serve as a **reference** for the **development team**.

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- 2. **Design** - A **design** is a **design** that is **developed** for the **purpose** of **meeting** the **requirements** of the **customer**.
- 3. **Manufacture** - A **manufacture** is a **manufacture** that is **developed** for the **purpose** of **meeting** the **requirements** of the **customer**.

5. **Requirements** - The requirements for the **product** are as follows:

- 1. **Performance** - The **product** must be able to **perform** at a **level** of **performance** that is **consistent** with the **requirements** of the **customer**.
- 2. **Reliability** - The **product** must be able to **reliably** **perform** at a **level** of **performance** that is **consistent** with the **requirements** of the **customer**.
- 3. **Cost** - The **product** must be able to **perform** at a **level** of **performance** that is **consistent** with the **requirements** of the **customer**.

6. **Conclusion** - This document is a **technical specification** for the **development** of a **new product**. It is intended to provide a **clear and concise** description of the **requirements** for the **product** and to serve as a **reference** for the **development team**.

1. A rectangular area is to be enclosed by a fence. The area is to be divided into two equal parts by a fence parallel to one of the sides. The total length of the fence is 1000 ft. Find the dimensions of the rectangle that will give the maximum area.

Solution:

Let x be the length of the rectangle and y be the width. The area is $A = xy$. The perimeter is $P = 2x + 2y = 1000$. The area is $A = xy = x\left(\frac{1000 - 2x}{2}\right) = 500x - x^2$. The area is a function of x : $A(x) = 500x - x^2$. The area is a downward-opening parabola. The maximum area occurs at $x = \frac{-b}{2a} = \frac{-500}{-2} = 250$. The width is $y = \frac{1000 - 2x}{2} = \frac{1000 - 500}{2} = 250$. The dimensions are 250 ft by 250 ft.



2. A rectangular area is to be enclosed by a fence. The area is to be divided into two equal parts by a fence parallel to one of the sides. The total length of the fence is 1000 ft. Find the dimensions of the rectangle that will give the maximum area.

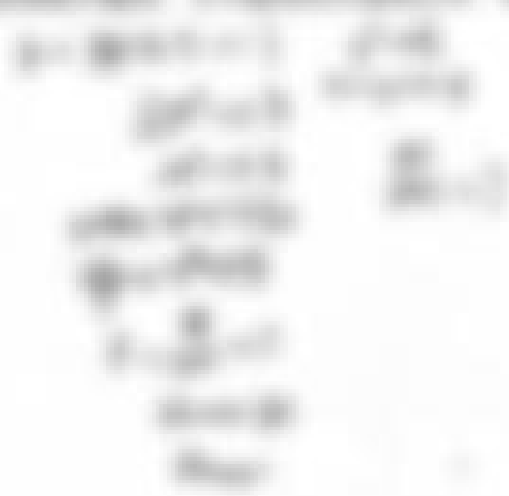
3. A rectangular area is to be enclosed by a fence. The area is to be divided into two equal parts by a fence parallel to one of the sides. The total length of the fence is 1000 ft. Find the dimensions of the rectangle that will give the maximum area.

Solution:

Let x be the length of the rectangle and y be the width. The area is $A = xy$. The perimeter is $P = 2x + 2y = 1000$. The area is $A = xy = x\left(\frac{1000 - 2x}{2}\right) = 500x - x^2$. The area is a function of x : $A(x) = 500x - x^2$. The area is a downward-opening parabola. The maximum area occurs at $x = \frac{-b}{2a} = \frac{-500}{-2} = 250$. The width is $y = \frac{1000 - 2x}{2} = \frac{1000 - 500}{2} = 250$. The dimensions are 250 ft by 250 ft.

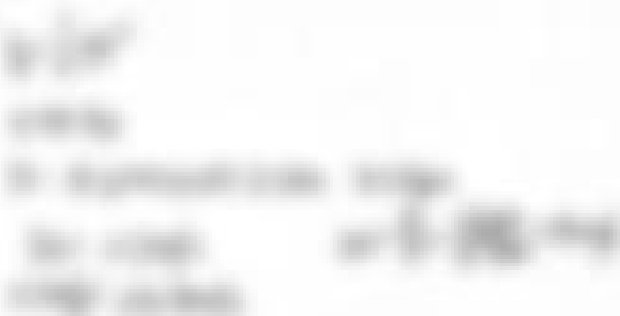


1. The following information is given for the reaction of hydrogen gas with chlorine gas to form hydrogen chloride gas. The reaction is exothermic.



2. The following information is given for the reaction of hydrogen gas with chlorine gas to form hydrogen chloride gas. The reaction is exothermic.

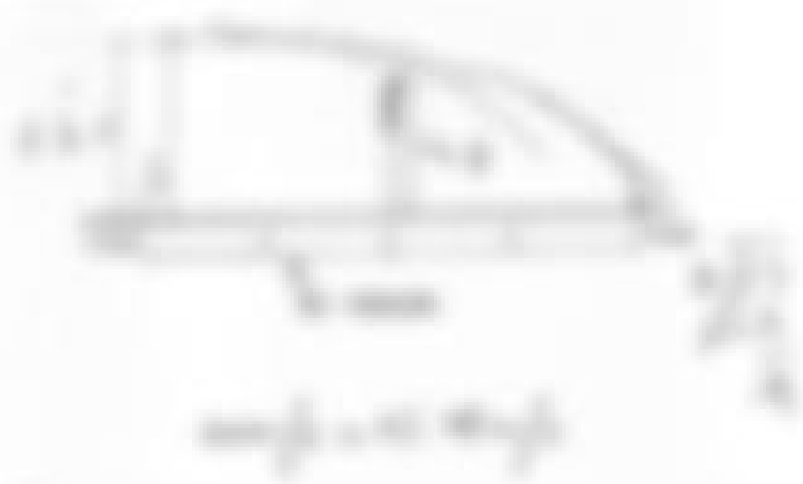
- (i) Enthalpy change of formation of $HCl(g)$ is $-92.3 \text{ kJ mol}^{-1}$.
- (ii) Enthalpy change of formation of $H_2O(l)$ is $-285.8 \text{ kJ mol}^{-1}$.



1. **QUESTION:** A rectangular block of material is subjected to a uniform stress σ in the x -direction. The block has a length L in the x -direction, a width b in the y -direction, and a height h in the z -direction. The material has a Young's modulus E . Determine the change in length ΔL of the block.

ANSWER: The change in length ΔL is given by:

$$\Delta L = \frac{\sigma L}{E}$$



2. **QUESTION:** A rectangular block of material is subjected to a uniform stress σ in the y -direction. The block has a length L in the x -direction, a width b in the y -direction, and a height h in the z -direction. The material has a Young's modulus E . Determine the change in width Δb of the block.

ANSWER: The change in width Δb is given by:



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1. **RESEARCH DESIGN**—This study was a descriptive, cross-sectional study. The data were collected from a convenience sample of 100 nurses working in a tertiary care hospital in the city of Pune, India. The study was approved by the Institutional Review Board of the hospital.

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1. **QUESTION** (10 marks): A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the top edge of the block, perpendicular to the vertical face. Determine the minimum value of F required to tip the block over the edge.

2. **ANSWER** (10 marks):

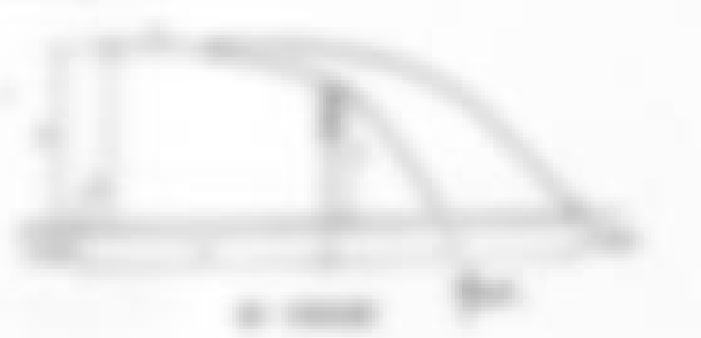
Solution:

Let the block be represented by a rectangle of width b and height h . The center of mass is at the geometric center, at a horizontal distance of $\frac{b}{2}$ from the left edge and a vertical distance of $\frac{h}{2}$ from the bottom edge.

When the block is about to tip, it is in equilibrium. The forces acting on it are:

- Weight $W = mg$ acting downwards from the center of mass.
- Normal reaction N acting upwards at the pivot point (the bottom-left corner).
- Applied force F acting horizontally to the right at the top-left corner.

For the block to be in equilibrium, the sum of the moments about the pivot point must be zero.



3. **QUESTION** (10 marks): A uniform rod of length L and mass M is pivoted at one end. A force F is applied at the other end, perpendicular to the rod. Determine the angular acceleration of the rod.

4. **ANSWER** (10 marks):

Let the rod be pivoted at point O . The force F is applied at point A , which is at a distance L from O . The force is perpendicular to the rod.

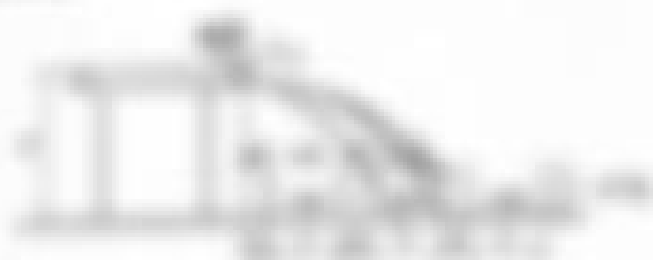
The torque τ about the pivot is given by:

$$\tau = F \times L$$

The moment of inertia I of the rod about the pivot is:

$$I = \frac{1}{3}ML^2$$

Using Newton's second law for rotation, $\tau = I\alpha$, we can find the angular acceleration α :

$$F \times L = \left(\frac{1}{3}ML^2\right)\alpha$$
$$\alpha = \frac{3F}{ML}$$


5. **QUESTION** (10 marks): A particle of mass m moves in a circular path of radius r with a constant speed v . Determine the centripetal force acting on the particle.

6. **ANSWER** (10 marks):

The centripetal force F_c is given by:

$$F_c = \frac{mv^2}{r}$$

where m is the mass of the particle, v is its speed, and r is the radius of the circular path.

1. A rectangular block of mass m and height h is placed on a horizontal surface. A force F is applied to the top right corner of the block, directed at an angle θ to the horizontal. The block is on the verge of tipping clockwise about its bottom-left corner. Determine the magnitude of the force F in terms of m , h , and θ .

2. A uniform rod of length L and mass M is pivoted at its left end. A force F is applied to the right end of the rod, directed at an angle ϕ to the rod. The rod is in equilibrium in a horizontal position. Determine the magnitude of the force F in terms of M , L , and ϕ .

3. A uniform rod of length L and mass M is pivoted at its left end. A force F is applied to the right end of the rod, directed at an angle ϕ to the rod. The rod is in equilibrium in a horizontal position. Determine the magnitude of the force F in terms of M , L , and ϕ .



4. A uniform rod of length L and mass M is pivoted at its left end. A force F is applied to the right end of the rod, directed at an angle ϕ to the rod. The rod is in equilibrium in a horizontal position. Determine the magnitude of the force F in terms of M , L , and ϕ .

5. A uniform rod of length L and mass M is pivoted at its left end. A force F is applied to the right end of the rod, directed at an angle ϕ to the rod. The rod is in equilibrium in a horizontal position. Determine the magnitude of the force F in terms of M , L , and ϕ .

6. A uniform rod of length L and mass M is pivoted at its left end. A force F is applied to the right end of the rod, directed at an angle ϕ to the rod. The rod is in equilibrium in a horizontal position. Determine the magnitude of the force F in terms of M , L , and ϕ .



1. **Introduction** - **Background** - **Objectives** - **Scope** - **Limitations** - **Assumptions** - **Definitions** - **Notations** - **References** - **Appendices** - **Index** - **Summary** - **Conclusions** - **Recommendations** - **Future Work** - **Acknowledgments** - **Disclaimer** - **Conflicts of Interest** - **Author Contributions** - **Data Availability Statement** - **Supplementary Materials** - **References** - **Appendices** - **Index** - **Summary** - **Conclusions** - **Recommendations** - **Future Work** - **Acknowledgments** - **Disclaimer** - **Conflicts of Interest** - **Author Contributions** - **Data Availability Statement** - **Supplementary Materials**



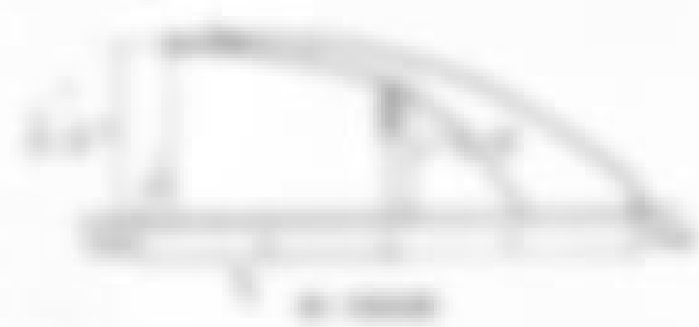
2. **Methodology** - **Research Design** - **Data Collection** - **Data Analysis** - **Results** - **Discussion** - **Conclusion** - **References** - **Appendices** - **Index** - **Summary** - **Conclusions** - **Recommendations** - **Future Work** - **Acknowledgments** - **Disclaimer** - **Conflicts of Interest** - **Author Contributions** - **Data Availability Statement** - **Supplementary Materials** - **References** - **Appendices** - **Index** - **Summary** - **Conclusions** - **Recommendations** - **Future Work** - **Acknowledgments** - **Disclaimer** - **Conflicts of Interest** - **Author Contributions** - **Data Availability Statement** - **Supplementary Materials**

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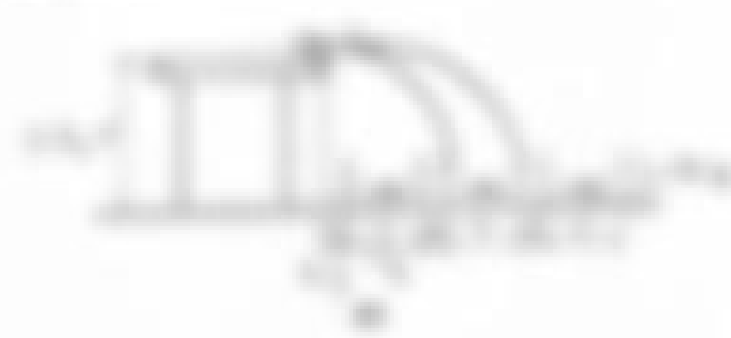
1. The following information is given for the structure shown in the figure. The structure is a frame with a fixed support at A and a roller support at B. The frame consists of a horizontal member AB of length 10 m and a vertical member BC of height 4 m. A uniformly distributed load of 2 kN/m is applied vertically downwards over the entire length of member AB. The frame is subjected to a horizontal point load of 10 kN acting to the right at joint B. The frame is also subjected to a clockwise moment of 20 kNm at joint B. The frame is supported by a fixed support at A and a roller support at B. The frame is shown in the figure.

Draw the shear force and bending moment diagrams for the frame.



2. The following information is given for the structure shown in the figure. The structure is a frame with a fixed support at A and a roller support at B. The frame consists of a horizontal member AB of length 10 m and a vertical member BC of height 4 m. A uniformly distributed load of 2 kN/m is applied vertically downwards over the entire length of member AB. The frame is subjected to a horizontal point load of 10 kN acting to the right at joint B. The frame is also subjected to a clockwise moment of 20 kNm at joint B. The frame is supported by a fixed support at A and a roller support at B. The frame is shown in the figure.

- a. Draw the shear force and bending moment diagrams for the frame.
- b. Calculate the maximum shear force and bending moment in the frame.



Handwritten notes: $P = 10 \text{ kN}$

1. A rectangular channel with a width of 10 ft and a depth of 4 ft is flowing at a velocity of 10 ft/s. The channel is made of concrete with a roughness coefficient of 0.012. Determine the head loss due to friction over a length of 100 ft.

2. A trapezoidal channel with a bottom width of 10 ft, side slopes of 1:1, and a depth of 4 ft is flowing at a velocity of 10 ft/s. The channel is made of concrete with a roughness coefficient of 0.012. Determine the head loss due to friction over a length of 100 ft.

- 3. A circular pipe with a diameter of 10 ft is flowing at a velocity of 10 ft/s. The pipe is made of concrete with a roughness coefficient of 0.012. Determine the head loss due to friction over a length of 100 ft.
- 4. A rectangular channel with a width of 10 ft and a depth of 4 ft is flowing at a velocity of 10 ft/s. The channel is made of concrete with a roughness coefficient of 0.012. Determine the head loss due to friction over a length of 100 ft.



5. A rectangular channel with a width of 10 ft and a depth of 4 ft is flowing at a velocity of 10 ft/s. The channel is made of concrete with a roughness coefficient of 0.012. Determine the head loss due to friction over a length of 100 ft.

- 6. A trapezoidal channel with a bottom width of 10 ft, side slopes of 1:1, and a depth of 4 ft is flowing at a velocity of 10 ft/s. The channel is made of concrete with a roughness coefficient of 0.012. Determine the head loss due to friction over a length of 100 ft.
- 7. A circular pipe with a diameter of 10 ft is flowing at a velocity of 10 ft/s. The pipe is made of concrete with a roughness coefficient of 0.012. Determine the head loss due to friction over a length of 100 ft.

- 8. A rectangular channel with a width of 10 ft and a depth of 4 ft is flowing at a velocity of 10 ft/s. The channel is made of concrete with a roughness coefficient of 0.012. Determine the head loss due to friction over a length of 100 ft.
- 9. A trapezoidal channel with a bottom width of 10 ft, side slopes of 1:1, and a depth of 4 ft is flowing at a velocity of 10 ft/s. The channel is made of concrete with a roughness coefficient of 0.012. Determine the head loss due to friction over a length of 100 ft.



1. Definition: A function $f: D \rightarrow \mathbb{R}$ is called odd if $f(-x) = -f(x)$ for all $x \in D$.

Properties: If f is odd, then $f(0) = 0$.

- Examples:
- $f(x) = x^3$ is odd.
 - $f(x) = x^2$ is even.
 - $f(x) = \sin(x)$ is odd.
 - $f(x) = \cos(x)$ is even.
 - $f(x) = e^x$ is neither odd nor even.
- Graph: The graph of an odd function is symmetric with respect to the origin.



2. Definition: A function $f: D \rightarrow \mathbb{R}$ is called even if $f(-x) = f(x)$ for all $x \in D$.

Properties: If f is even, then $f(0)$ is defined.

- Examples:
- $f(x) = x^2$ is even.
 - $f(x) = x^4$ is even.
 - $f(x) = \cos(x)$ is even.
 - $f(x) = e^{x^2}$ is even.
- Graph: The graph of an even function is symmetric with respect to the y-axis.



Graphs:

- $f(x) = x^3$ (odd function)
- $f(x) = x^2$ (even function)
- $f(x) = \sin(x)$ (odd function)
- $f(x) = \cos(x)$ (even function)

Conclusion: The graph of an odd function is symmetric with respect to the origin, and the graph of an even function is symmetric with respect to the y-axis.

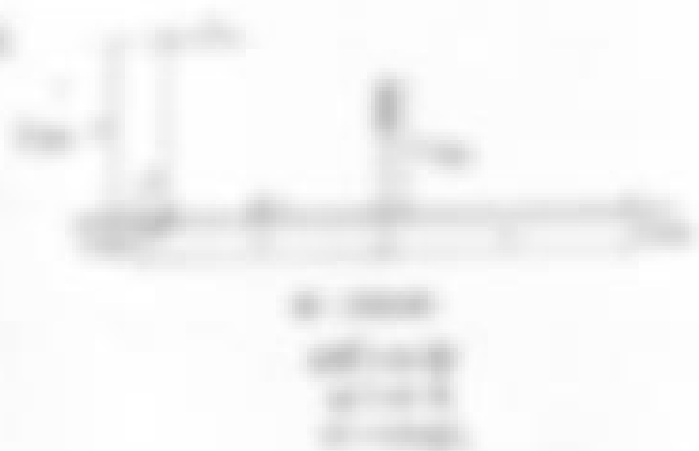
1. **QUESTION** (10 marks) – A beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A uniformly distributed load of 2 kN/m is applied over the entire length of the beam. Calculate the reaction forces at the supports.

SOLUTION (10 marks) –

Let R_L be the reaction force at the left support and R_R be the reaction force at the right support. The total load is $W = 2 \times 10 = 20$ kN. The reaction forces are given by:

$$R_L + R_R = W$$
$$R_L = 20 - R_R$$

Since the beam is in equilibrium, the sum of the moments about the left support is zero:

$$R_R \times 10 - W \times 5 = 0$$
$$R_R = \frac{W \times 5}{10} = \frac{20 \times 5}{10} = 10 \text{ kN}$$
$$R_L = 20 - 10 = 10 \text{ kN}$$


2. **QUESTION** (10 marks) – A beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A point load of 10 kN is applied at the center of the beam. Calculate the reaction forces at the supports.

SOLUTION (10 marks) –

Let R_L be the reaction force at the left support and R_R be the reaction force at the right support. The reaction forces are given by:

$$R_L + R_R = 10$$

Since the beam is in equilibrium, the sum of the moments about the left support is zero:

$$R_R \times 10 - 10 \times 5 = 0$$
$$R_R = \frac{10 \times 5}{10} = 5 \text{ kN}$$
$$R_L = 10 - 5 = 5 \text{ kN}$$

3. **QUESTION** (10 marks) – A beam of length 10 m is supported by a pin support at the left end and a roller support at the right end. A point load of 10 kN is applied at the center of the beam. Calculate the reaction forces at the supports.



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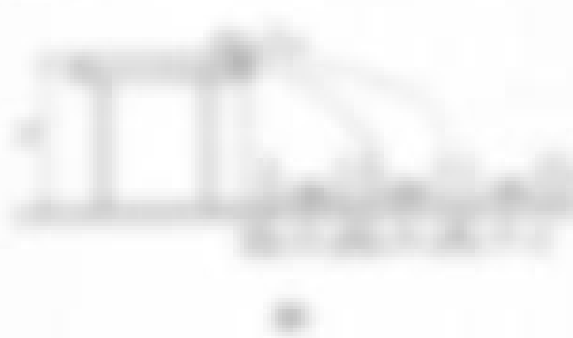
1. **RESEARCH DESIGN**—This study was a descriptive, cross-sectional study. The data were collected from a convenience sample of 100 nurses working in a tertiary care hospital in the city of Pune, India. The study was approved by the Institutional Review Board of the hospital. The data were collected from January to March 2015.

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1. The following information is given for the structure shown in the figure. The structure is a frame with a horizontal beam of length 10 m and a vertical column of height 4 m. The beam is supported by a pin support at the left end and a roller support at the right end. The column is supported by a pin support at the base. A uniformly distributed load of 2 kN/m is applied vertically downwards over the entire length of the beam. A point load of 10 kN is applied vertically downwards at the free end of the beam. The angle of inclination of the beam with the horizontal is 30°.



2. The following information is given for the structure shown in the figure. The structure is a frame with a horizontal beam of length 10 m and a vertical column of height 4 m. The beam is supported by a pin support at the left end and a roller support at the right end. The column is supported by a pin support at the base. A uniformly distributed load of 2 kN/m is applied vertically downwards over the entire length of the beam. A point load of 10 kN is applied vertically downwards at the free end of the beam. The angle of inclination of the beam with the horizontal is 30°.



1. **QUESTION** (10 marks) A right-angled triangle has a square inscribed in it. The side of the square is 4 cm. Find the area of the triangle.

SOLUTION (10 marks) Let the side of the square be x cm.

a. $\triangle ABC$ is a right-angled triangle.
The side of the square is 4 cm.



$$\begin{aligned} \text{Area of } \triangle ABC &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times (4 + 8) \times 12 \end{aligned}$$

$$= \frac{1}{2} \times 12 \times 12 = 72 \text{ cm}^2$$

2. **QUESTION** (10 marks) A right-angled triangle has a square inscribed in it. The side of the square is 4 cm. Find the area of the triangle.

SOLUTION (10 marks) Let the side of the square be x cm.

a. $\triangle ABC$ is a right-angled triangle. The side of the square is 4 cm.

b. $\triangle ABC$ is a right-angled triangle.



$$\begin{aligned} \text{Area of } \triangle ABC &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times (4 + 8) \times 12 \end{aligned}$$

$$= \frac{1}{2} \times 12 \times 12 = 72 \text{ cm}^2$$

c. $\triangle ABC$ is a right-angled triangle.

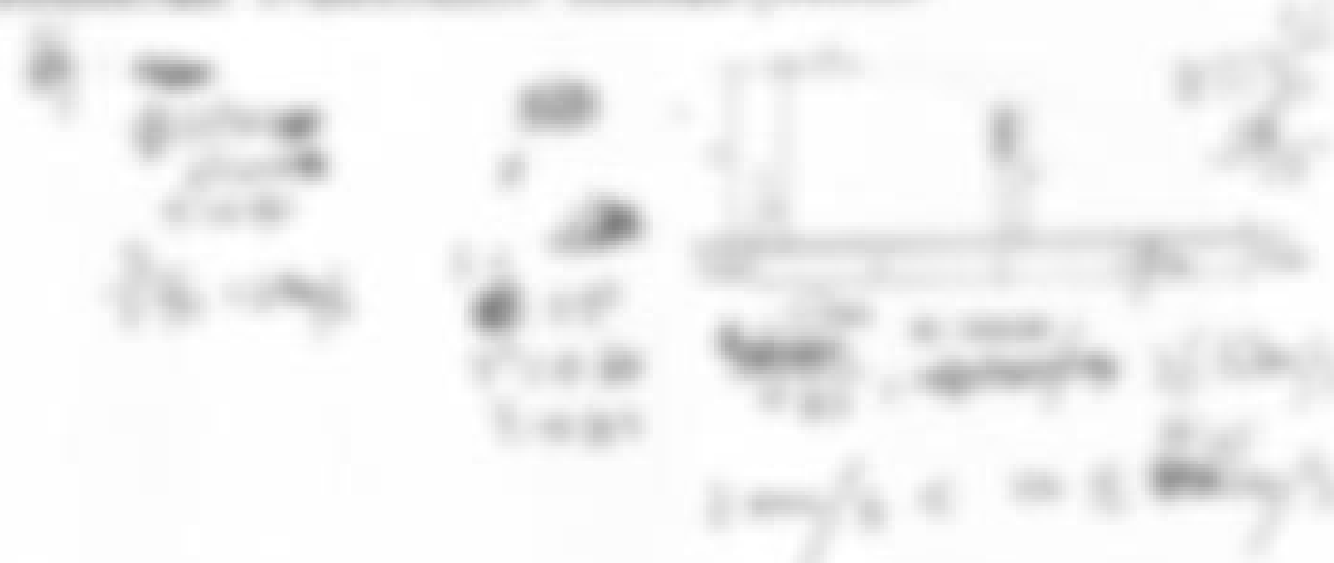
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Journal of Internal Medicine 247: 399–406

1. **Abstract.** *Abstract text describing the paper's content.*

1. $\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{1}{2} m \frac{d}{dt} (v^2)$

1. A rectangular block of material is shown in the diagram. The block is 10 cm long, 5 cm wide and 2 cm high. The block is made of a material with a density of 2.5 g/cm³. Calculate the mass of the block.

2. A rectangular block of material is shown in the diagram. The block is 10 cm long, 5 cm wide and 2 cm high. The block is made of a material with a density of 2.5 g/cm³. Calculate the mass of the block.



3. A rectangular block of material is shown in the diagram. The block is 10 cm long, 5 cm wide and 2 cm high. The block is made of a material with a density of 2.5 g/cm³. Calculate the mass of the block.

4. A rectangular block of material is shown in the diagram. The block is 10 cm long, 5 cm wide and 2 cm high. The block is made of a material with a density of 2.5 g/cm³. Calculate the mass of the block.



1. **QUESTION** (10 marks)

Figure 1 shows a simply supported beam of length 10 m. The beam is subjected to a uniformly distributed load of 2 kN/m over its entire length. The beam is supported by a pin support at A and a roller support at B. The beam is divided into three equal segments of 3.33 m each. The beam is shown in Figure 1.



Figure 1

2. **ANSWER** (10 marks)

The beam is shown in Figure 1. The beam is subjected to a uniformly distributed load of 2 kN/m over its entire length. The beam is supported by a pin support at A and a roller support at B. The beam is divided into three equal segments of 3.33 m each. The beam is shown in Figure 1.

1. **QUESTION** (10 marks)

Figure 2 shows a simply supported beam of length 10 m. The beam is subjected to a uniformly distributed load of 2 kN/m over its entire length. The beam is supported by a pin support at A and a roller support at B. The beam is divided into three equal segments of 3.33 m each. The beam is shown in Figure 2.



Figure 2

2. **ANSWER** (10 marks)

The beam is shown in Figure 2. The beam is subjected to a uniformly distributed load of 2 kN/m over its entire length. The beam is supported by a pin support at A and a roller support at B. The beam is divided into three equal segments of 3.33 m each. The beam is shown in Figure 2.

1. **PROBLEM STATEMENT:** A rectangular block of material is subjected to a uniform stress σ_x in the x -direction. The block has a height h and a width b . The stress σ_x is applied to the vertical faces of the block.

ASSUMPTIONS: The material is homogeneous and isotropic. The stress σ_x is uniform across the entire cross-section of the block.

Let σ_x be the stress in the x -direction. The stress σ_x is applied to the vertical faces of the block. The block has a height h and a width b . The stress σ_x is uniform across the entire cross-section of the block.

Let σ_y be the stress in the y -direction. The stress σ_y is applied to the horizontal faces of the block. The block has a height h and a width b . The stress σ_y is uniform across the entire cross-section of the block.

Let σ_z be the stress in the z -direction. The stress σ_z is applied to the faces of the block perpendicular to the z -axis. The block has a height h and a width b . The stress σ_z is uniform across the entire cross-section of the block.



2. **PROBLEM STATEMENT:** A rectangular block of material is subjected to a uniform stress σ_y in the y -direction. The block has a height h and a width b . The stress σ_y is applied to the horizontal faces of the block.

ASSUMPTIONS: The material is homogeneous and isotropic. The stress σ_y is uniform across the entire cross-section of the block.

Let σ_x be the stress in the x -direction. The stress σ_x is applied to the vertical faces of the block. The block has a height h and a width b . The stress σ_x is uniform across the entire cross-section of the block.



•

1. **QUESTION** (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with initial speed u at an angle α to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point A . The time taken for the particle to reach A is T . The horizontal distance between O and A is R . The vertical height of the particle at any time t is y . The horizontal distance of the particle from O at any time t is x . The particle is at a height y at two different times t_1 and t_2 . The horizontal distance of the particle from O at these two times is x_1 and x_2 respectively. The particle is at a height y at two different times t_1 and t_2 . The horizontal distance of the particle from O at these two times is x_1 and x_2 respectively.

Answer: (10 marks)

$u \sin \alpha = \frac{y}{t_1} = \frac{y}{t_2}$
 $u \cos \alpha = \frac{x_1}{t_1} = \frac{x_2}{t_2}$
 $\frac{y}{t_1} = \frac{y}{t_2}$
 $\frac{x_1}{t_1} = \frac{x_2}{t_2}$
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 $\frac{y}{x_1} = \frac{y}{x_2}$



or $y = \frac{1}{2} g t^2$

2. **QUESTION** (10 marks)
A particle of mass m is projected from the origin O of a Cartesian coordinate system with initial speed u at an angle α to the horizontal. The particle moves in a parabolic path and strikes the horizontal axis at a point A . The time taken for the particle to reach A is T . The horizontal distance between O and A is R . The vertical height of the particle at any time t is y . The horizontal distance of the particle from O at any time t is x . The particle is at a height y at two different times t_1 and t_2 . The horizontal distance of the particle from O at these two times is x_1 and x_2 respectively. The particle is at a height y at two different times t_1 and t_2 . The horizontal distance of the particle from O at these two times is x_1 and x_2 respectively.

Answer: (10 marks)

$u \sin \alpha = \frac{y}{t_1} = \frac{y}{t_2}$
 $u \cos \alpha = \frac{x_1}{t_1} = \frac{x_2}{t_2}$
 $\frac{y}{t_1} = \frac{y}{t_2}$
 $\frac{x_1}{t_1} = \frac{x_2}{t_2}$
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 $\frac{y}{x_1} = \frac{y}{x_2}$
 $\frac{y}{x_1} = \frac{y}{x_2}$



or $y = \frac{1}{2} g t^2$



1. A rectangular area is divided into four smaller rectangles by a horizontal line and a vertical line. The dimensions of the rectangles are given in the table below.

Find the area of the shaded rectangle.

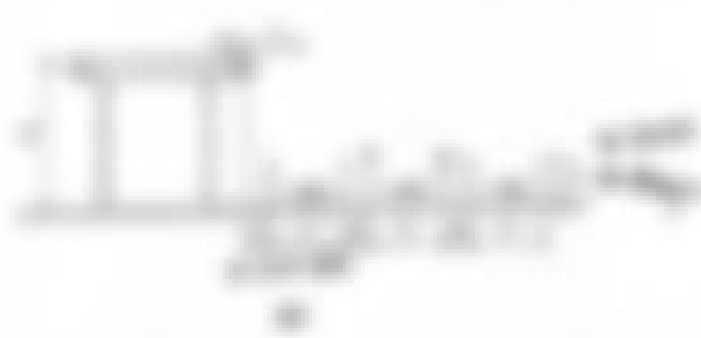
- Length = 10 cm
- Width = 8 cm
- Area of rectangle A = 20 cm²
- Area of rectangle B = 16 cm²
- Area of rectangle C = 12 cm²
- Area of rectangle D = 8 cm²



2. A rectangular area is divided into four smaller rectangles by a horizontal line and a vertical line. The dimensions of the rectangles are given in the table below.

Find the area of the shaded rectangle.

- Length = 10 cm
- Width = 8 cm
- Area of rectangle A = 20 cm²
- Area of rectangle B = 16 cm²
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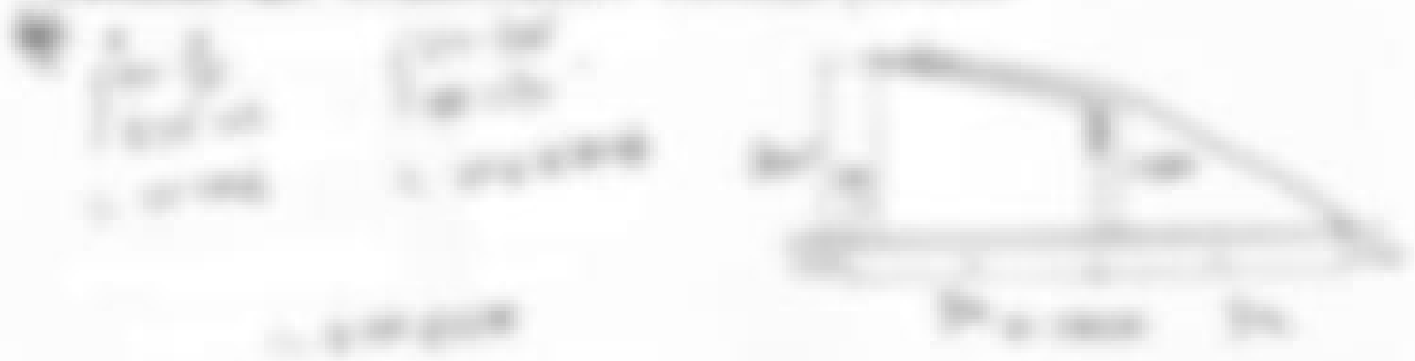
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1. **QUESTION** (10 marks)
A simply supported beam of length 10 m is subjected to a uniformly distributed load of 2 kN/m over its entire length. The beam has a rectangular cross-section with a width of 0.2 m and a height of 0.4 m. The material has a Young's modulus of $E = 200 \text{ GPa}$. Determine the maximum deflection of the beam.

SOLUTION (10 marks)



2. **QUESTION** (10 marks)
A simply supported beam of length 10 m is subjected to a point load of 10 kN at its midpoint. The beam has a rectangular cross-section with a width of 0.2 m and a height of 0.4 m. The material has a Young's modulus of $E = 200 \text{ GPa}$. Determine the maximum deflection of the beam.

SOLUTION (10 marks)

3. **QUESTION** (10 marks)
A simply supported beam of length 10 m is subjected to a triangular load with a maximum intensity of 2 kN/m at its midpoint. The beam has a rectangular cross-section with a width of 0.2 m and a height of 0.4 m. The material has a Young's modulus of $E = 200 \text{ GPa}$. Determine the maximum deflection of the beam.



1. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom. The block is on the verge of tipping over the right edge. Find the magnitude of the force F .

① $\sum \tau = 0$ (about the right edge)

$\tau_{\text{gravity}} = \tau_{\text{force}}$

$mgh = Fy$

$F = \frac{mgh}{y}$

② $\sum F_x = 0$ and $\sum F_y = 0$

$F = mg$ (if $y = h$)

③ $\sum \tau = 0$ (about the left edge)

$\tau_{\text{gravity}} = \tau_{\text{normal}}$

$mgh = N \cdot \frac{h}{2}$

$N = 2mg$

④ $\sum F_x = 0$ and $\sum F_y = 0$

$F = N$

$F = 2mg$



2. A rectangular block of mass m and height h is placed on a horizontal surface. A horizontal force F is applied to the block at a height y from the bottom. The block is on the verge of tipping over the right edge. Find the magnitude of the force F .

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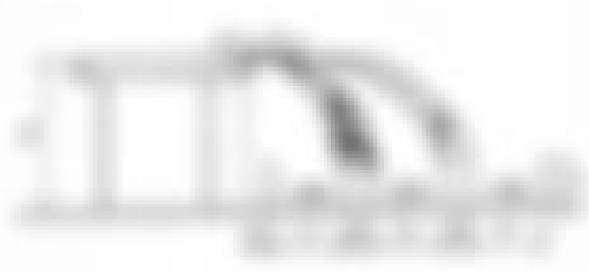
1. The first part of the document is a list of the names of the persons who have been appointed to the various positions of the Board of Directors of the Corporation. The names are listed in alphabetical order, and each name is followed by the position to which he or she has been appointed. The list is as follows:

2. The second part of the document is a list of the names of the persons who have been appointed to the various positions of the Board of Directors of the Corporation. The names are listed in alphabetical order, and each name is followed by the position to which he or she has been appointed. The list is as follows:

3. The third part of the document is a list of the names of the persons who have been appointed to the various positions of the Board of Directors of the Corporation. The names are listed in alphabetical order, and each name is followed by the position to which he or she has been appointed. The list is as follows:

4. The fourth part of the document is a list of the names of the persons who have been appointed to the various positions of the Board of Directors of the Corporation. The names are listed in alphabetical order, and each name is followed by the position to which he or she has been appointed. The list is as follows:

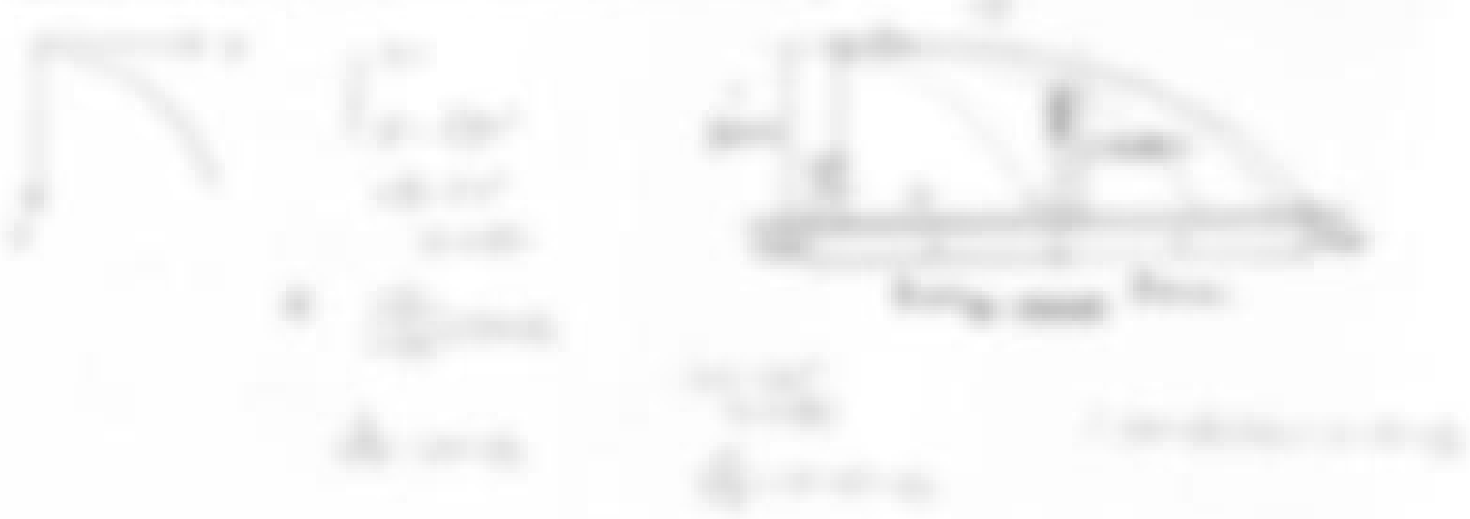
5. The fifth part of the document is a list of the names of the persons who have been appointed to the various positions of the Board of Directors of the Corporation. The names are listed in alphabetical order, and each name is followed by the position to which he or she has been appointed. The list is as follows:



3.

1. A rectangular plate of width b and height h is subjected to a uniformly distributed load w acting downwards. The plate is supported by a pin support at the left end and a roller support at the right end. Determine the reactions at the supports.

2. A rectangular plate of width b and height h is subjected to a uniformly distributed load w acting downwards. The plate is supported by a pin support at the left end and a roller support at the right end. Determine the reactions at the supports.



3. A rectangular plate of width b and height h is subjected to a uniformly distributed load w acting downwards. The plate is supported by a pin support at the left end and a roller support at the right end. Determine the reactions at the supports.

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5. A rectangular plate of width b and height h is subjected to a uniformly distributed load w acting downwards. The plate is supported by a pin support at the left end and a roller support at the right end. Determine the reactions at the supports.



Answer/ans

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1. *Journal of Management Education*, 2000, 24(1), 10-12. © 2000 Sage Publications, Inc. 10.1177/002203210024001010

1. *Journal of the American Medical Association*, 1998; 279: 1033-1037.

